Memory-Augmented Autoencoder With Adaptive Reconstruction and Sample Attribution Mining for Hyperspectral Anomaly Detection

Yu Huo, Xi Cheng, *Student Member, IEEE*, Sheng Lin, Min Zhang, *Member, IEEE*, and Hai Wang

***Abstract*— Hyperspectral anomaly detection (HAD) aims to identify targets that are significantly different from their surrounding background, employing an unsupervised paradigm. Recently, detectors based on autoencoder (AE) have become predominant methods and demonstrated satisfactory performance. However, there are still two problems that need to be solved. First, the hypothesis that the AE-based models can effectively reconstruct background samples (BSs) while anomalies cannot, may not always be true in practice, due to their powerful capability for feature extraction. Second, the AE-based models primarily concentrate on the quality of sample reconstruction, regardless of whether the encoded features signify the anomalies or background, which is not conducive to the separation of anomalies from the background. To handle the above-mentioned problems, a novel memory-augmented AE (MAAE) model is developed to better reconstruct the background and suppress anomalies reconstruction. Specifically, for the first problem, a novel superpixel-guided adaptive weight calculation (SAWC) module is devised to generate adaptive weights (AWs) by taking into account contextual information in the error map, and then the AWs are incorporated into the reconstruction loss, where the potential BSs are endowed with larger AWs than anomalies during training. For the second problem, a novel sample attribution mining (SAM) module is developed to mine sample attribution (i.e., explore whether a certain sample belongs to the background or anomaly), and the mined background and anomaly samples (ASs) are employed to train different modules for better separating the anomalies and background. In addition, an entropy-based sparse addressing (ESA) module is further designed to weaken the reconstruction ability of ASs by designing a learnable sparse addressing weight for the memory module. The ablation study validates the effectiveness of the proposed SAWC, SAM, and ESA. Extensive comparison experiments on six hyperspectral image (HSI) datasets demonstrate the superiority in terms of comprehensive detection performance and background suppression of our method.**

***Index Terms*— Adaptive reconstruction, entropy-based sparse addressing (ESA), hyperspectral anomaly detection (HAD), memory-augmented autoencoder (MAAE), sample attribution mining (SAM).**

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The authors are with the School of Aerospace Science and Technology, Xidian University, Xi'an 710126, China (e-mail: yuhuo@stu.xidian.edu.cn; cheng_xi@stu.xidian.edu.cn; linsheng@stu.xidian.edu.cn; minzhang@xidian.edu.cn; wanghai@mail.xidian.edu.cn).

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I. Introduction

HYPERSPECTRAL images (HSIs) provide rich spectral information, making it possible to accurately distinguish ground materials [1], [2], [3], [4], [5], and have received significant attention in the domain of remote sensing research. Among them, the hyperspectral anomaly detection (HAD) task in HSIs is with great significance for hyperspectral information processing and can be applied to environmental monitoring [6], [7], intelligent agriculture [8], [9], and military affairs [10], [11]. The HAD task seeks anomalous targets that exhibit spectral characteristics significantly distinct from their neighboring background, utilizing an unsupervised manner.

With the advancement of deep learning technology [12], [13], [14], [15], [16], reconstruction-based HAD approaches have demonstrated satisfactory performance. Among them, the autoencoder (AE)-based method is one of the typical works, and its theoretical basis is that the model can effectively reconstruct background samples (BSs), while anomalies cannot. Consequently, the anomalies can be detected straightforwardly through the reconstruction error. However, there are still two open problems that need to be solved in AE-based HAD methods. First, the hypothesis that AE-based models can effectively reconstruct BSs while anomalies cannot, may not always be true in practice, due to their powerful capability for feature extraction. Consequently, both anomalies and background exhibit small reconstruction errors, and the detection results derived from such errors may lead to many anomalies not being detected, thereby reducing the HAD model performance. Second, the AE-based models primarily concentrate on the quality of sample reconstruction, regardless of whether the encoded features signify the anomalies or background. Actually, the model is expected to have a better reconstruction ability for the background rather than anomalies. When anomalies and background are fed to the same model for training, it simultaneously improves the reconstruction ability for both anomalies and background. Consequently, it is not conducive to the separation of anomalies and backgrounds and limits the upper bound of HAD performance.

To handle the above-mentioned problems, a novel memory-augmented AE (MAAE) model is developed to better reconstruct the background and suppress anomalies reconstruction for HAD in HSIs, as shown in Fig. 1. Specifically, to handle the first problem, a novel superpixel-guided adaptive

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weight calculation (SAWC) module is proposed to generate adaptive weights (AWs) by taking into account contextual information in the error map (i.e., the errors between original and reconstructed data), which is then incorporated into the reconstruction loss, focusing the training on BSs while dampening the emphasis on anomaly samples (ASs). Specifically, the AWs are first calculated by mapping superpixel segmentation regions to the reconstruction error map, the details can be seen in Section III-B. Then, AWs are incorporated into the reconstruction loss to adaptively attend to samples with different attributions, in which the potential BSs are endowed with larger AWs than anomalies during training. Consequently, the MAAE is beneficial for reconstructing BSs but is not conducive to the reconstruction of abnormal ones.

To handle the second problem, a novel sample attribution mining (SAM) module is proposed to mine sample attribution (i.e., explore whether a certain sample belongs to the background or anomaly), in which mined pseudo or potential BSs are employed to train encoder, memory, and decoder modules, and the pseudo-ASs are utilized to train encoder and decoder modules. The motivation is to improve the ability of the memory module to memorize BSs while ignoring the memorization of ASs. Consequently, the memory module can suppress the interference of ASs mixed in the training samples, thereby improving the separation degree between anomalies and background. To alleviate the issue that ASs may be fairly reconstructed well when using dense addressing weights. Entropy-based sparse addressing (ESA) module is further designed to obtain sparse addressing weight for the memory module, which is utilized to weaken the reconstruction ability for abnormal samples.

In this article, three modules are designed to handle the aforementioned problems. Specifically, the design motivation for the first module, SAWC, is to allocate large weights to the BSs to focus on their learning during training. The design motivation for the second module, SAM, is to prevent the training of ASs in the memory module as much as possible. Although assisted by the above two modules, some anomalies might still have the opportunity to be effectively reconstructed with a sophisticated combination of memory items through a dense addressing weight containing numerous elements [17]. Therefore, the design motivation for the third module, ESA, is to encourage sparsity in the addressing weights, thereby weakening the model's ability to reconstruct ASs. In summary, we handled the aforementioned problems by focusing on the learning of BSs, preventing the memory of ASs, and weakening the reconstruction of ASs. The primary contributions of this article are encapsulated as follows.

1) A novel SAWC module is developed to generate AWs by taking into account the contextual information in the error map, and the AWs are then incorporated into the reconstruction loss to adaptively reconstruct samples, in which the potential BSs are endowed with larger AWs than anomalies during training. Consequently, the MAAE is beneficial for reconstructing BSs but is not conducive to the reconstruction of abnormal ones.
2) A novel SAM module is devised to explore sample attribution, and solely the mined pseudo or potential BSs are utilized to train the memory module. The aim is to enhance the ability of the memory module to memorize BSs while ignoring the memorization of abnormal samples. Therefore, the memory module can suppress the interference of abnormal samples mixed in the training samples.
3) To further reduce the chance of reconstructing anomalies, an ESA module is devised to weaken the model's reconstruction ability to ASs by designing a learnable sparse addressing weight for the memory module. Consequently, collaborating the SAM module with the ESA module can enhance the separation degree between anomalies and background, thereby improving the detection performance.

II. Related Work

In recent years, more and more excellent HAD methods [6], [18], [19] have been proposed. This section introduces some classic anomaly detection (AD) methods, including statistic-based, representation-based, reconstruction-based, and some other classic works.

A. Statistic-Based HAD Methods

The statistical-based HAD methods typically build an assessment model for the background to identify anomalies. Among them, the Reed–Xiaoli (RX) [20] is a representative algorithm, it assumes that the background follows a Gaussian distribution, and the anomalies are determined by calculating the Mahalanobis distance between the test sample and the distribution. After this, numerous improved versions have been proposed to build more credible background estimation models. For example, a local RX (LRX) [21] method was proposed, which employs local window pixels to precisely estimate background distribution. A weighted RX (WRX) detector [22] was developed to better estimate the background information by assigning AWs for anomaly and BSs. Kwon and Nasrabadi [23] devised a kernel RX (KRX) detector, which nonlinearly projects spectral features into a high-dimensional space, and achieves more effective spectrum separation. Zhou et al. [24] designed an improved KRX detector, which can efficiently reduce the algorithm complexity via employing a clustering strategy that comprehensively considers both global and local background information. Schaum [25] devised a subspace RX (SSRX) model to better estimate background distribution by reducing the impact of anomalies or noisy pixels.

B. Representation-Based HAD Methods

The theoretical foundation of representation-based methods lies in the concept that the BSs can be represented effectively via several atoms in an overcomplete dictionary, while anomalies cannot. For collaborative representation, Li and Du [26] developed a representative work, termed the collaborative representation detector (CRD), which posits central pixels can be effectively represented based on their neighboring pixels, whereas anomalies cannot. Vafadar and Ghassemian [27]

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proposed an improved CRD by removing outliers that are substantially distinct from the background within a dual window. For sparsity representation, Li et al. [28] integrated sparse representation with a triple-window model to structure a purer background dictionary. Yuan et al. [29] introduced the concept of local sparse divergence to detect abnormal pixels. The pixels with higher sparsity divergence are identified as abnormal pixels. For low-rank representation, Xu et al. [30] devised a low-rank and sparse representation (LRASR) model that can better characterize the representation of each pixel by integrating a sparsity-inducing component into the representation coefficient. Li et al. [31] introduced a low-rank and sparse decomposition model with a mixture of Gaussian (LSDM-MOG), aiming to effectively describe the intricate distribution by integrating a mixture noise model with the low-rank background. Other classical works, such as [32], [33], [34], [35], similarly, combined LRASR to improve the HAD performance.

C. Reconstruction-Based HAD Methods

The theoretical basis of reconstruction-based methods is that the model can effectively reconstruct background pixels, while anomalies, due to their very small proportion relative to background pixels, cannot be reconstructed well. The reconstruction-based HAD methods are categorized into two primary categories: AE network and generative adversarial network (GAN).

For the AE-based networks, Bati et al. [36] first introduced the AE network into the HAD, which can reconstruct the background and adapt to the varying complexity of HSIs. Based on this idea, many methods have been proposed in the literature. For example, Wang et al. [37] developed an autonomous HAD (Auto-AD) network to construct a more pure background by incorporating the designed adaptive-weighted into reconstruction loss to suppress the reconstruction of abnormal pixels. Fan et al. [38] designed a robust graph AE (RGAE) model by integrating a graph regularization term into AE to maintain consistency of geometric structure and local space, while decreasing the time consumption. Liu et al. [39] devised a dual-frequency AE detector, which transformed the HSI into high- and low-frequency components to improve the separability of the anomalies and background. Zhao and Sun [40] introduced a novel memory AE (MAE) network that incorporates a memory module into AE to inhibit the reconstruction of anomaly pixels.

For the GAN-based networks, Jiang et al. [41] constructed a spectral-constrained GAN with a weakly supervised learning paradigm, enhancing the distinction between background and anomalies to improve model performance. Li et al. [42] developed a sparse coding-inspired GAN, which can effectively learn distinction reconstruction with low errors for background pixels and large errors for anomaly pixels. Jiang et al. [43] introduced a semi-supervised learning manner into the GAN, which can effectively enhance the background modeling accuracy and contribute to model performance. Another classic GAN-based approach [44], [45], similarly, integrated the GAN model into HAD to improve HAD performance.

In addition, some other classic AD works have been proposed. For example, Díaz et al. [46] designed a line-by-line fast anomaly detector (LbL-FAD) that detects anomalies by automatically evaluating the threshold of the background distribution. Zhang and Cheng [47] developed an unsupervised transferred convolutional neural network, which can effectively extract spatial and spectral information by employing a tensor structure. Cheng et al. [48] proposed a deep features-based two-stream isolation forest network for HAD, which can sufficiently utilize the spectral and spatial characteristics of HSI to improve the model performance. Coca et al. [49] developed a deep convolutional network to detect anomalies, which utilizes a self-supervised strategy to learn the image characteristics. Li et al. [50] devised a general HAD model that requires training only once, and the model can learn the spatial context features between background and anomalies by an unsupervised paradigm.

III. Proposed Method

The overall framework of the proposed MAAE is illustrated in Fig. 1, and it is constructed on the MAE benchmark by incorporating the proposed three modules including Fig. 1(b) ESA, Fig. 1(c) SAM, and Fig. 1(d) SAWC to better reconstruct the background and suppress anomaly reconstruction for HAD in HSIs. To be specific, the SAWC module is developed to generate AWs, which are incorporated into the reconstruction loss to focus training on BSs while dampening the emphasis on ASs. The SAM module is devised to explore whether a certain sample belongs to the background or anomaly according to the AWs calculated by the SAWC module, and then solely the mined pseudo or potential BSs are imported into the memory module to enhance the ability of the memory module to memorize BSs while ignoring the memorization of abnormal ones. In addition, an ESA module is further designed to obtain sparse addressing weight, which is utilized to further weaken the reconstruction ability of ASs. The details of the MAAE are as follows.

A. Preliminary

1) AE Framework: The AE models are typically used for modeling high-dimensional data in an unsupervised manner. It comprises an encoder module and a decoder module. The encoder is utilized to encode the original data into a low-dimensional representation. The decoder is employed to reconstruct original data from the low-dimensional representation. Specifically, let $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, $\mathbf{Z} \in \mathbb{R}^{H\times W\times C}$ (To simplify the demonstration, it is unfolded into 2-D data in the figure), and $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times B}$ indicate original data, latent representation, and reconstructed data, respectively, where H and W separately indicate corresponding height and width, B and C indicate corresponding band numbers. Given the ith sample $\mathbf{x}_i \in \mathbf{X}$, the encoder converts it to latent representation $\mathbf{z}_i \in \mathbf{Z}$, and the decoder is utilized to map $\mathbf{z}_i$ to reconstructed data $\hat{\mathbf{x}}_i \in \hat{\mathbf{X}}$. The details are as follows:

$$\mathbf{z}_i = f_e(\mathbf{x}_i) \tag{1}$$

$$\hat{\mathbf{x}}_i = f_d(\mathbf{z}_i) \tag{2}$$

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where $f_e(\cdot)$ and $f_d(\cdot)$ indicate encoder and decoder, respectively. Finally, the AE model is optimized using reconstruction loss L_R

$$L_R = \left\|\mathbf{X} - \hat{\mathbf{X}}\right\|_2 \tag{3}$$

where $\|\cdot\|_2$ represents L_2 norm.

2) MAE Framework: The traditional MAE model is constructed by incorporating a memory module into the AE model, which comprises an encoder module, a memory module, and a decoder module. The MAE process is as follows. The encoder is first utilized to encode the original data into a latent representation. Then, different from the AE model, the MAE does not directly import the latent representation into the decoder but uses it as a query to retrieve the most relevant memory items in the memory module. Among them, the memory module comprises a memory matrix and a memory addressing mechanism. The memory matrix $\mathbf{M} \in \mathbb{R}^{K\times C}$ is designed as a learnable matrix, which contains K vectors with C dimension. Specifically, given a query (i.e., latent representation) $\mathbf{z}_i$, the ith retrieved memory item $\hat{\mathbf{z}}_i \in \hat{\mathbf{Z}}$ can be obtained via the following equation:

$$\hat{\mathbf{z}}_i = \mathbf{w}_i\mathbf{M} = \sum_{k=1}^{K} w_{ik}\mathbf{m}_k \tag{4}$$

where $\mathbf{m}_k \in \mathbb{R}^{1\times C}$ represents the kth row memory item in $\mathbf{M}$, $\mathbf{w}_i \in \mathbb{R}^{1\times K}$ denotes the ith row addressing weight vector in $\mathbf{W}$, and w_{ik} denotes the kth addressing weight in $\mathbf{w}_i$, which is obtained via implementing memory addressing operator as follows:

$$w_{ik} = \frac{\mathbf{z}_i\mathbf{m}_k^T}{\|\mathbf{z}_i\|_1\left\|\mathbf{m}_k^T\right\|_1} \tag{5}$$

where $\mathbf{z}_i \in \mathbb{R}^{1\times C}$ represents the ith row query in $\mathbf{Z}$, superscript T represents matrix transpose, $\|\cdot\|_1$ represents L_1 norm.

Finally, the retrieved matrix items obtained from (4) are input into the decoder to obtain the reconstructed data. It should be noted that the encoder and decoder in the MAE have the same framework as those in the AE.

B. SAWC for Decoder Module

The AWs, denoted as $\mathbf{W}^a$, is acquired by the SAWC module, as shown in Fig. 1(d). Specifically, the initial error value $\mathrm{ie}_{i,j}$ at position (i, j) is first calculated through the following equation:

$$\mathrm{ie}_{i,j} = \sum_{b=1}^{B}\left|x_{b,i,j} - \hat{x}_{b,i,j}\right| \tag{6}$$

where $x_{b,i,j}$ expresses the pixel value of the original HSI $\mathbf{X}$ for the bth band at position (i, j), $\hat{x}_{b,i,j}$ expresses the pixel value of reconstructed HSI $\hat{\mathbf{X}}$ for the same band and position. The initial error map, denoted as $\mathrm{IE} \in \mathbb{R}^{H\times W}$, can be obtained by calculating the initial error value $\mathrm{ie}_{i,j}$ of each pixel. Second, the superpixel segmentation method is employed to obtain superpixel segmentation regions. Specifically, the first principal component (FPC) of HSI is first acquired by using the method of principal component analysis (PCA), then the simple linear iterative cluster (SLIC) algorithm [51] is utilized to divide the FPC into multiple nonoverlapping superpixel segmentation regions $R = \{r_1, r_2, \ldots, r_n, \ldots, r_{N_R}\}$, where N_R denotes the number of superpixel segmentation regions. Third, the $E_R = \{E_{r_1}, E_{r_2}, \ldots, E_{r_n}, \ldots, E_{rN}\}$ is acquired through mapping the R to IE, where $E_{r_n} \in \mathbb{R}^{H\times W}$ is the subset of E_R, which is obtained by padding zero around r_n. The illustration of E_{r_n} can be seen in Fig. 2(a). Fourth, the fine error map FE_{r_n} of E_{r_n} is calculated by taking into account contextual reconstruction errors as follows:

$$\mathrm{FE}_{r_n} = \begin{cases} \dfrac{1}{8}\displaystyle\sum_{k=1}^{8}\left|\mathrm{ie}_k - \mathrm{ie}_{r_n}\right|, & \text{if } \mathrm{ie}_{r_n} \in r_n \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where ie_{r_n} denotes a certain element value in E_{r_n}, and ie_k is the element value of adjacent to ie_{r_n}. Note that the value of ie_k is equal to 0 when $\mathrm{ie}_k \notin r_n$.

Fifth, the final fine error map $\mathrm{FE}_R \in \mathbb{R}^{H\times W}$ can be acquired through combining each FE_{r_n}, which is formulated as follows:

$$\mathrm{FE}_R = \mathrm{FE}_{r_1} \oplus \mathrm{FE}_{r_2} \oplus \cdots \oplus \mathrm{FE}_{r_n} \oplus \cdots \oplus \mathrm{FE}_{r_N} \tag{8}$$

where $\oplus$ denotes element-wise addition operation. In the end, the AW $\mathbf{W}^a$ is calculated as follows:

$$\mathbf{W}^a = e^{-\alpha\cdot\mathrm{FE}_R} \tag{9}$$

where α is a hyperparameter used for controlling the focus degree to BSs.

The design motivation for the AWs $\mathbf{W}^a$ is based on the following fact and assumption [37]: the background is predominant in an HSI, while anomalies have a low probability of occurrence. Therefore, this is an assumption that the extreme imbalance in the number of anomalies and BSs results in anomalies having larger reconstruction errors than the background during training.

Based on the above fact and assumption, if the ie_{r_n} denotes the error of a BS, the $\mathbf{W}^a$ associated with the position of ie_{r_n} has a large value. If the ie_{r_n} denotes the error of an AS, the $\mathbf{W}^a$ associated with the position of ie_{r_n} has a small value. It can be confirmed by Fig. 2(c), where the white region in $\mathbf{W}^a$ denotes large weight values associated with the background, and the black points in $\mathbf{W}^a$ denote small weight values associated with the anomalies. To give a more intuitive comparison, the original error weight map and the AW map are demonstrated in Fig. 2(b) and (c), respectively. Observations show that the AW map can assign more precise weights to the background and ASs compared to the original error weight map during training.

C. SAM and ESA for Memory Module

1) Sample Attribution Mining: The SAM module is devised to explore whether a certain sample belongs to the background or anomaly, and the mined pseudo/potential BS and pseudo-ASs are trained using different training strategies to suppress the interference of abnormal samples mixed in the training samples. The details of SAM are as follows.

1) *Pseudo BS Mining:* As shown in Fig. 1(c), first, sort the values of $\mathbf{W}^a$ in descending order. Next, select

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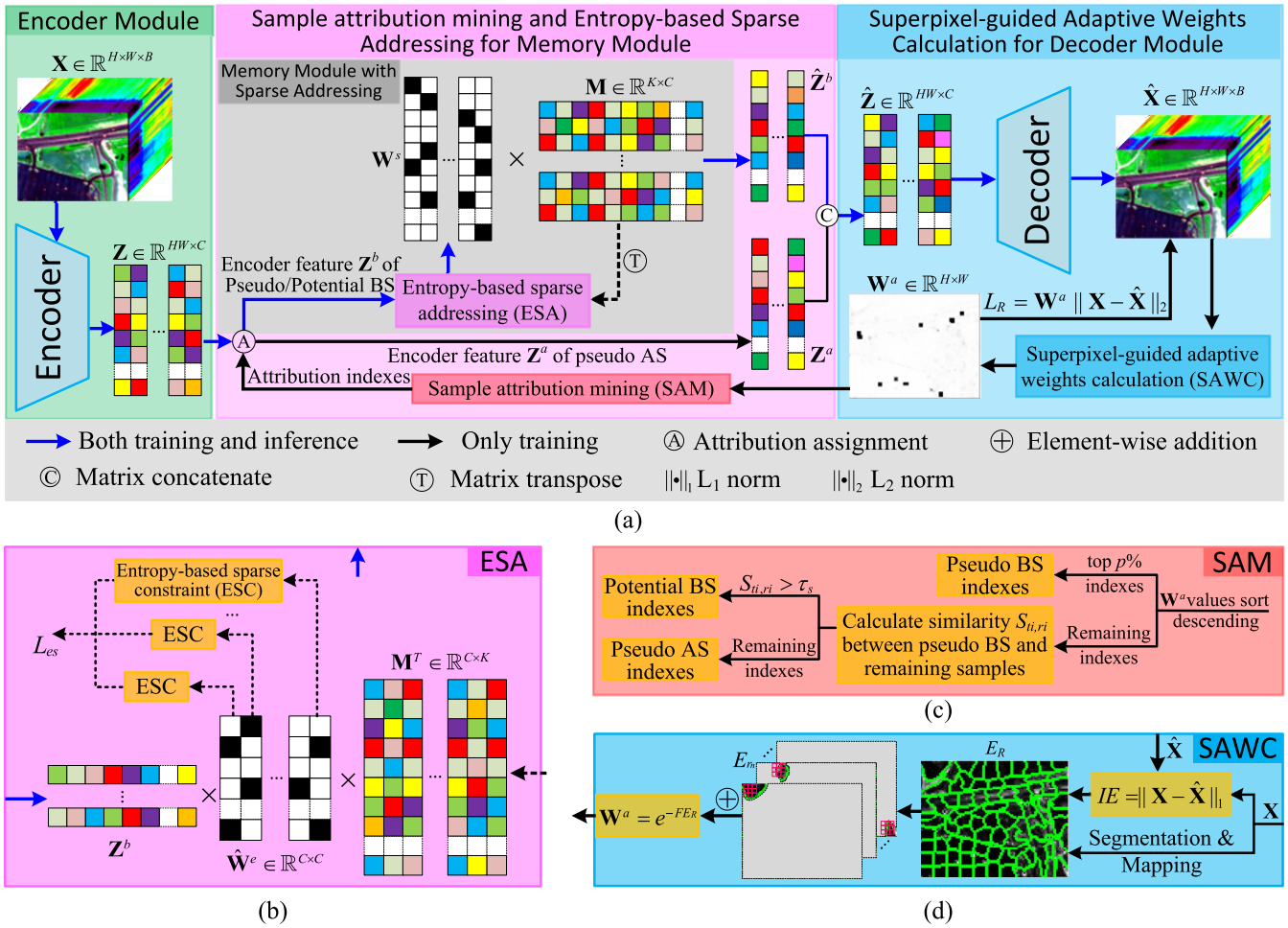


Fig. 1. (a) Overall framework of MAAE is constructed on the MAE benchmark by incorporating the proposed three modules including (b) ESA, (c) SAM, and (d) SAWC.

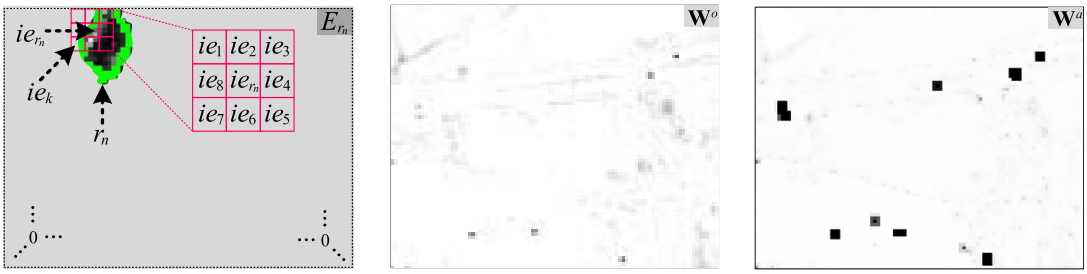


Fig. 2. Illustration of (a) E_{r_n}, (b) $\mathbf{W}^o$, and (c) $\mathbf{W}^a$, where E_{r_n} denotes the error map corresponding to the superpixel segmentation region r_n, $\mathbf{W}^o$ and $\mathbf{W}^a$ denote the original error weight map and the AW map, respectively.

top p percent indexes $\mathrm{TI} = \{\mathrm{ti}_1, \mathrm{ti}_2, \ldots, \mathrm{ti}_m, \ldots, \mathrm{ti}_{M_{\mathrm{TI}}}\}$, whose corresponding samples are regarded as pseudo BS $\mathbf{X}_{\mathrm{TI}}$, where M_{TI} denotes the number of top p percent indexes. The remaining indexes and their corresponding remaining samples (RSs) are represented as $\mathrm{RI} = \{\mathrm{ri}_1, \mathrm{ri}_2, \ldots, \mathrm{ri}_n, \ldots, \mathrm{ri}_{N_{\mathrm{RI}}}\}$ and $\mathbf{X}_{\mathrm{RI}}$, respectively, where N_{RI} denotes the number of remaining indexes.

2) *Potential BS and Pseudo AS Mining:* The similarity graph is built to search potential BS from $\mathbf{X}_{\mathrm{RI}}$. Specifically, a similarity graph is defined by $G^s = (V^s, E^s)$, where each node in V^s indicates a searched potential BS which has high similarity with the pseudo-BS, and each edge in E^s denotes the similarity between pseudo-BS $\mathbf{X}_{\mathrm{TI}}$ and RS $\mathbf{X}_{\mathrm{RI}}$. The initial similarity, denoted as $\mathrm{IS}_{\mathrm{ti}_m,\mathrm{ri}_n}$, can be acquired by calculating the spectral angular distance between the $\mathbf{X}_{\mathrm{TI}}$ and $\mathbf{X}_{\mathrm{RI}}$, which is formulated as follows:

$$\mathrm{IS}_{\mathrm{ti}_m,\mathrm{ri}_n} = \arccos\left(\frac{\mathbf{x}_{\mathrm{ti}_m}\mathbf{x}_{\mathrm{ri}_n}}{\left\|\mathbf{x}_{\mathrm{ti}_m}\right\|_2\left\|\mathbf{x}_{\mathrm{ri}_n}\right\|_2}\right) \tag{10}$$

where $\mathbf{x}_{\mathrm{ti}_m} \in \mathbf{X}_{\mathrm{TI}}$ denotes one of the pseudo BS, $\mathbf{x}_{\mathrm{ri}_n} \in \mathbf{X}_{\mathrm{RI}}$ denotes one of the RS.

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The set of $\mathrm{IS}_{\mathrm{ti}_m,\mathrm{ri}_n}$ is defined as $\mathbf{IS}_{\mathrm{TI,RI}} = \{\mathrm{IS}_{\mathrm{ti}_1,\mathrm{ri}_1}, \mathrm{IS}_{\mathrm{ti}_1,\mathrm{ri}_2}, \ldots, \mathrm{IS}_{\mathrm{ti}_{M_{\mathrm{TI}}},\mathrm{ri}_{N_{\mathrm{RI}}}}\}$, and the final similarity $S_{\mathrm{ti}_m,\mathrm{ri}_n}$ is normalized to the range $[0, 1]$ through the following transformation:

$$S_{\mathrm{ti}_m,\mathrm{ri}_n} = \frac{\mathrm{IS}_{\mathrm{ti}_m,\mathrm{ri}_n} - \mathrm{Min}\{\mathbf{IS}_{\mathrm{TI,RI}}\}}{\mathrm{Max}\{\mathbf{IS}_{\mathrm{TI,RI}}\} - \mathrm{Min}\{\mathbf{IS}_{\mathrm{TI,RI}}\}} \tag{11}$$

where $\mathrm{Max}\{\cdot\}$ and $\mathrm{Min}\{\cdot\}$ separately denote taking the maximum and minimum values from the set $\mathbf{IS}_{\mathrm{TI,RI}}$. If the sample $\mathbf{x}_{\mathrm{ri}_n}$ meets the condition that $S_{\mathrm{ti}_m,\mathrm{ri}_n} > \tau_s$, such the sample will be appended to G^s as potential BS. Consequently, we greedily mine all may potential BS into G^s as long as the $\mathbf{x}_{\mathrm{ri}_n}$ meets the above condition, and the rest of samples that have not been mined will be defined as pseudo-AS.

2) Entropy-Based Sparse Addressing: As shown in Fig. 1(b), a learnable sparse addressing weight $\mathbf{W}^s$ is devised to measure the similarity between the query $\mathbf{Z}^b$ and the memory items $\mathbf{M} \in \mathbb{R}^{K\times C}$. The $w^s_{ik} \in \mathbf{W}^s$ is calculated as follows:

$$w^s_{ik} = \frac{\mathbf{z}_i \hat{\mathbf{W}}^e \mathbf{m}_k^T}{\|\mathbf{z}_i\|_1 \left\|\mathbf{m}_k^T\right\|_1} \tag{12}$$

where $\mathbf{z}_i \in \mathbb{R}^{1\times C}$ represents the ith row query in $\mathbf{Z}^b$, $\mathbf{m}_k \in \mathbb{R}^{1\times C}$ represents the kth row memory item in $\mathbf{M}$, $\hat{\mathbf{W}}^e \in \mathbb{R}^{C\times C}$ denotes a learnable matrix. The details of $\hat{\mathbf{W}}^e$ are as follows.

First of all, the matrix $\mathbf{W}^e \in \mathbb{R}^{C\times C}$ is initialized using uniform distribution. Second, the $\hat{\mathbf{W}}^e = \{\hat{\mathbf{w}}^e_1, \hat{\mathbf{w}}^e_2, \ldots, \hat{\mathbf{w}}^e_c, \ldots, \hat{\mathbf{w}}^e_C\}$ is obtained by normalizing $\mathbf{W}^e = \{\mathbf{w}^e_1, \mathbf{w}^e_2, \ldots, \mathbf{w}^e_c, \ldots, \mathbf{w}^e_C\}$ along column direction using the softmax operation

$$\hat{\mathbf{w}}^e_c = \left\{ \frac{e^{w^e_{cj}}}{\sum_{i=1}^{C} e^{w^e_{ci}}} \right\}_{j=1}^{C} \tag{13}$$

where w^e_{ci} and w^e_{cj} separately denote the ith and jth element in $\mathbf{w}^e_c$. Third, an entropy-based sparse constraint (ESC) is implemented via minimizing the entropy of each $\hat{\mathbf{w}}^e_c$

$$E\left(\hat{\mathbf{w}}^e_c\right) = -\sum_{i=1}^{C} \hat{w}^e_{ci} \log\left(\hat{w}^e_{ci}\right). \tag{14}$$

The ith retrieved memory item $\hat{\mathbf{z}}_i \in \hat{\mathbf{Z}}$ is acquired through the following equation:

$$\hat{\mathbf{z}}_i = \mathbf{w}^s_i \mathbf{M} = \sum_{k=1}^{K} w^s_{ik} \mathbf{m}_k \tag{15}$$

where $\mathbf{w}^s_i \in \mathbb{R}^{1\times K}$ denotes the ith row sparse addressing weight in $\mathbf{W}^s$, and w^s_{ik} calculated in (12) denotes the kth sparse addressing weight in $\mathbf{w}^s_i$.

D. Training and Inference

1) Model Training: The proposed MAAE is demonstrated as shown in Fig. 1. The whole training process is bifurcated into two stages, i.e., pretraining and training phase.

Algorithm 1 Training Process of MAAE

Input: Hyperspectral image $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$.
  for each epoch **do**
    if *epoch* ≤ 200 **then**
      Step 1: Feed $\mathbf{X}$ into encoder to obtain encoded feature $\mathbf{Z}$;
      Step 2: Obtain the final encoded feature $\hat{\mathbf{Z}}$ by implementing $\hat{\mathbf{Z}} \leftarrow \mathbf{Z}$, then feed $\hat{\mathbf{Z}}$ into decoder to obtain reconstructed hyperspectral image $\hat{\mathbf{X}}$;
      Step 3: Calculate adaptive weight $\mathbf{W}^a$ using Eq. (9);
      Step 4: Train encoder and decoder by minimizing L_R in Eq. (16).
    else
      Step 1: Feed $\mathbf{X}$ into encoder to obtain encoded feature $\mathbf{Z}$;
      Step 2: Feed $\mathbf{W}^a$ into SAM module to determine the pseudo/potential BS (i.e., $\mathbf{Z}^b$) and pseudo AS (i.e., $\mathbf{Z}^a$);
      Step 3: Feed $\mathbf{Z}^a$ into ESA module to obtain sparse addressing weight $\mathbf{W}^s$;
      Step 4: Calculate the retrieved memory item $\hat{\mathbf{Z}}^b$ according to $\mathbf{W}^s$ by using Eq. (15);
      Step 5: Concatenate $\mathbf{Z}^a$ and $\hat{\mathbf{Z}}^b$ to obtain final encoded feature $\hat{\mathbf{Z}}$.
      Step 6: Feed $\hat{\mathbf{Z}}$ into decoder to obtain reconstructed hyperspectral image $\hat{\mathbf{X}}$;
      Step 7: Train encoder, decoder and memory module by minimizing L in Eq. (18).
    end if
  end for
Output: A trained MAAE model.

Algorithm 2 Inference Process of MAAE

Input: Hyperspectral image $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$.
  Step 1: Feed $\mathbf{X}$ into encoder to obtain encoded feature $\mathbf{Z}$;
  Step 2: Obtain encoded feature $\mathbf{Z}^b$ by implementing $\mathbf{Z}^b \leftarrow \mathbf{Z}$, then feed $\mathbf{Z}^b$ into memory module to obtain retrieved memory item $\hat{\mathbf{Z}}^b$;
  Step 3: Obtain final encoded feature $\hat{\mathbf{Z}}$ by implementing $\hat{\mathbf{Z}} \leftarrow \hat{\mathbf{Z}}^b$, then feed $\hat{\mathbf{Z}}$ into decoder to obtain reconstructed hyperspectral image $\hat{\mathbf{X}}$.
Output: Obtain detection result with Eq. (19).

a) Pretraining phase: During the pretraining phase, an HSI $\mathbf{X}$ is first fed into the encoder to obtain the encoded feature $\mathbf{Z}$. Second, the $\hat{\mathbf{Z}} = \mathbf{Z}^a = \mathbf{Z}$ is imported into the decoder to obtain reconstructed HSI $\hat{\mathbf{X}}$. Third, the SAWC module is implemented to obtain initial AW $\mathbf{W}^a$. Note that the SAM and ESA modules are not utilized, because the $\mathbf{W}^a$ is not enough reliable for mining sample attribution at the primary of training. Fourth, the loss function can be redefined by incorporating $\mathbf{W}^a$ into the traditional reconstruction loss defined in (3) in the form of weight in the pretraining phase

$$L_R = \mathbf{W}^a \left\|\mathbf{X} - \hat{\mathbf{X}}\right\|_2 = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} w^a_{i,j} \left\|\mathbf{x}_{i,j} - \hat{\mathbf{x}}_{i,j}\right\|_2 \tag{16}$$

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where L_R represents the proposed adaptive reconstruction loss, $\mathbf{x}_{i,j}$ denotes the vector at location (i, j) of the HSI $\mathbf{X}$, $\hat{\mathbf{x}}_{i,j}$ denotes the vector at location (i, j) of the reconstructed $\hat{\mathbf{X}}$, $w^a_{i,j}$ indicates the weight value at location (i, j) of the $\mathbf{W}^a$.

b) Training phase: During the training phase, the SAWC, SAM, and ESA modules are concurrently integrated into the proposed method. Specifically, an HSI $\mathbf{X}$ is first imported into the encoder to obtain the encoded feature $\mathbf{Z}$. Second, the initial AW $\mathbf{W}^a$ derived from the pretraining phase is fed into the SAM module to generate sample attribution indexes. These indexes are utilized to determine the pseudo/potential BS (i.e., query item, denoted as $\mathbf{Z}^b$) and pseudo-AS (denoted as $\mathbf{Z}^a$). Third, the query $\mathbf{Z}^b$ is sent to the ESA module to obtain sparse addressing weight $\mathbf{W}^s$, then the retrieved memory item $\hat{\mathbf{Z}}^b$ is acquired via (15). Fourth, the final encoded feature $\hat{\mathbf{Z}}$ is obtained through concatenating $\mathbf{Z}^a$ and $\hat{\mathbf{Z}}^b$, and the $\hat{\mathbf{Z}}$ is then fed into decoder to obtain reconstructed HSI $\hat{\mathbf{X}}$. Fifth, akin to steps three and four during the pretraining phase, the adaptive reconstruction loss L_R is also redefined by integrating the AW $\mathbf{W}^a$ into the traditional reconstruction loss. Meanwhile, an ESC loss L_{ESC} is derived by calculating each $E(\hat{\mathbf{w}}^e_c)$ defined in (14) as follows:

$$L_{\text{ESC}} = \frac{1}{C}\sum_{c=1}^{C} E\left(\hat{\mathbf{w}}^e_c\right). \tag{17}$$

Hence, the overall loss function for the training phase is formulated by the following equation:

$$L = L_R + \lambda L_{\text{ESC}} \tag{18}$$

where λ indicates the tradeoff parameter to balance the L_R and L_{ESC}. To provide a clearer description, the training process of MAAE is summarized in Algorithm 1.

2) Model Inference: During the inference, an HSI $\mathbf{X}$ is first fed into the encoder to obtain the encoded feature $\mathbf{Z}$. Second, the query $\mathbf{Z}^b = \mathbf{Z}$ is imported into the memory module to acquire the retrieved memory item $\hat{\mathbf{Z}}^b$. It should be noted that $\hat{\mathbf{Z}}$ is equal to $\hat{\mathbf{Z}}^b$ at this moment. Third, the $\hat{\mathbf{Z}}$ is fed into the decoder to obtain reconstructed HSI $\hat{\mathbf{X}}$. The blue line in Fig. 1(a) illustrates the inference process. Ultimately, the AD result is derived using the following equation:

$$r_{ij} = \left\|\mathbf{x}_{i,j} - \hat{\mathbf{x}}_{i,j}\right\|_2^2 \tag{19}$$

where r_{ij} denotes the detection result at location (i, j). To provide a clearer description, the inference process of MAAE is summarized in Algorithm 2.

IV. Experiments

A. Experiment Setup

1) Dataset Setup: Six benchmark datasets, including five real HSI datasets and one synthetic HSI dataset, are utilized to evaluate the performance of the proposed MAAE. The pseudo-color images and their corresponding ground truth (GT) maps are demonstrated as shown in Fig. 3. The specifics of these datasets are outlined below.

1) *Gainesville Dataset [52]:* The dataset was collected from the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor, covering the Gainesville city area. The dataset comprises an image array of 100 × 100 pixels, each with a spatial resolution of 3.5 m, and contains 191 spectral bands. Several ships on the water's surface are marked as anomalies.
2) *Los Angeles Dataset [53]:* The dataset was captured by the AVIRIS sensor, covering the Los Angeles area, USA. The dataset comprises an image array of 100 × 100 pixels, each with a spatial resolution of 7.1 m, and contains 205 spectral bands. The buildings with different sizes are labeled as anomalies.
3) *SpecTIR Dataset [54]:* The dataset was acquired through the SpecTIR hyperspectral airborne Rochester experiment (SHARE). The dataset comprises an image array of 180 × 180 pixels, each with a spatial resolution of 1 m, and contains 120 spectral bands. The man-made colorful square fabrics of varying sizes are marked as anomalies.
4) *Salinas Dataset [55]:* The dataset is a synthetic one, which is produced via executing the target implantation method [56] on a real dataset that was collected from the AVIRIS sensor, covering the Salinas Valley area in California, USA. It contains 120 × 120 pixels with 204 spectral bands, and its spatial resolution is 3.7 m per pixel.
5) *HYDICE Dataset [57]:* The dataset was captured using the Hyperspectral Digital Imagery Collection Experiment (HYDICE) airborne sensor, covering the urban area in California, USA. The dataset comprises an image array of 80 × 100 pixels, each with a spatial resolution of 1 m, and contains 175 spectral bands. Some roofs and cars with a total of 21 pixels are labeled as anomalies.
6) *San Diego Dataset [58]:* The dataset was collected from the AVIRIS sensor, covering the San Diego airport in California, USA. The dataset comprises an image array of 100 × 100 pixels, each with a spatial resolution of 3.5 m, and contains 189 spectral bands. Three airplanes are marked as anomalies.

For the Gainesville, Los Angeles, HYDICE, San Diego, and SpecTIR datasets, their GTs are manually labeled with the help of the Environment for Visualizing Images (ENVIs) software [52], [59]. The Salinas is a synthetic dataset, which is produced via executing the target implantation method [56] on a real dataset. Then, the RX detector is used to find pixels that are different from others and label them as anomaly targets [60].

2) Evaluation Metric: Extensive qualitative and quantitative evaluation metrics are used to evaluate the performance of our method on the six datasets mentioned above.

a) Qualitative evaluation: For the qualitative evaluation, a 3-D receiver operating characteristic (3-D ROC) curve [61] and an anomaly background separation map are utilized. Among them, three 2-D ROC curves relating to (P_D, P_F), (P_D, τ), and (P_F, τ) are obtained by holding one of the variables P_D (probability of detection), P_F (false alarm rate), and τ (threshold) in the 3-D ROC curve. The anomaly background separation map can visually demonstrate the separation level of the anomalies and background. The greater the disparity

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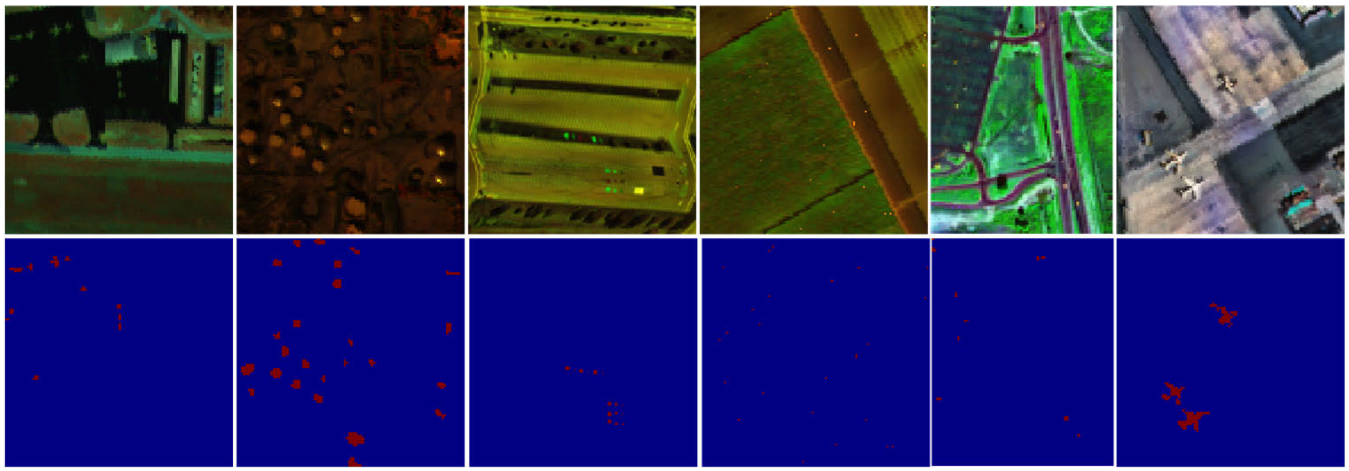


Fig. 3. First and second rows represent the pseudo-color images and their corresponding GT maps, respectively. (a) Gainesville. (b) Los Angeles. (c) SpecTIR. (d) Salinas. (e) HYDICE. (f) San Diego.

between the anomaly and background boxes, the more effective the separation.

b) Quantitative evaluation: For the quantitative evaluation, three area under the curve (AUC) values, including $\mathrm{AUC}(P_D, P_F)$, $\mathrm{AUC}(P_D, \tau)$, and $\mathrm{AUC}(P_F, \tau)$, are employed. Based on the three AUC values, another five AUC measures, including the target detectability ($\mathrm{AUC_{TD}}$), background suppressibility ($\mathrm{AUC_{BS}}$), overall detection probability ($\mathrm{AUC_{ODP}}$), target detectability with background suppressibility ($\mathrm{AUC_{TDBS}}$), and signal-to-noise probability ratio ($\mathrm{AUC_{SNPR}}$), can be further derived, which are defined as follows:

$$\begin{aligned}
\mathrm{AUC_{SNPR}} &= \frac{\mathrm{AUC}(P_D, \tau)}{\mathrm{AUC}(P_F, \tau)} \\
\mathrm{AUC_{TD}} &= \mathrm{AUC}(P_D, P_F) + \mathrm{AUC}(P_D, \tau) \\
\mathrm{AUC_{BS}} &= \mathrm{AUC}(P_D, P_F) - \mathrm{AUC}(P_F, \tau) \\
\mathrm{AUC_{TDBS}} &= \mathrm{AUC}(P_D, \tau) - \mathrm{AUC}(P_F, \tau) \\
\mathrm{AUC_{ODP}} &= \mathrm{AUC}(P_D, P_F) + \mathrm{AUC}(P_D, \tau) - \mathrm{AUC}(P_F, \tau)
\end{aligned} \tag{20}$$

where the higher the values of $\mathrm{AUC}(P_D, P_F)$, $\mathrm{AUC}(P_D, \tau)$, $\mathrm{AUC_{TD}}$, $\mathrm{AUC_{BS}}$, $\mathrm{AUC_{ODP}}$, $\mathrm{AUC_{TDBS}}$, and $\mathrm{AUC_{SNPR}}$, the better the model performance of the HAD is. The lower the value of $\mathrm{AUC}(P_F, \tau)$, the better the performance of background suppression is.

3) Implementation Details:

a) Model setting: The encoder module in MAAE is constructed using fully connected layers (FCLs), and node numbers of the FCLs are set to B, 128, 64, and 32, respectively, where B indicates the band numbers of the dataset. Similarly, the decoder is constructed using FCLs, and the node numbers of FCLs are set to 32, 64, 128, and B, respectively. Each FCL is followed by a batch normalization layer and a LeakyReLU function, and the last LeakyReLU function is replaced by the Sigmoid function to ensure the output is normalized between 0 and 1. In the ESA module, the size of the learnable matrix $\hat{\mathbf{W}}^e$ is set to 32×32. In the memory module, the size of the memory matrix $\mathbf{M}$ is set to $K \times 32$, The K is quantitatively analyzed in Section IV-B.

TABLE I
PARAMETER CONFIGURATION FOR THE MAAE MODEL ON SIX DATASETS

Datasets	λ	p (%)	N_R	LR	α	K
Gainesville	0.0001	50	50	0.0005	2	500
Los Angeles	0.0001	70	100	0.001	4	
SpecTIR	0.001	60	50	0.005	5	
Salinas	0.001	60	100	0.005	6	
HYDICE	0.0001	80	100	0.002	5	
San Diego	0.0001	80	50	0.001	5	

b) Training setting: The Adam optimizer [62] is used for the optimization of the proposed MAAE. The batch size corresponds to the number of samples in the dataset. The training epochs are set to 200 and 1800 during the pretraining and training phase, respectively. The threshold τ_s is set to 0.5. The parameters λ, top p, N_R, learning rate (LR), and α are quantitatively analyzed in Section IV-B.

c) Environment setting: All experiments are implemented on a system equipped with an Intel[1] Core[2] i7-9700 CPU running at 3.00 GHz and having 48 GB of RAM.

B. Parameter Analysis

1) Parameter Analysis of λ: The parameter λ is quantitatively analyzed on the six datasets. As shown in Fig. 4(a), the λ is configured as $\{0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}\}$. Observations show that the $\mathrm{AUC}(P_D, P_F)$ values exhibit a progressive increase when λ changes from 0 to 10^{-4}, with the highest values being reached at $\lambda = 10^{-4}$ on the Gainesville, Los Angeles, HYDICE, and San Diego datasets. The $\mathrm{AUC}(P_D, P_F)$ values exhibit a progressive increase when λ changes from 0 to 10^{-3}, with the highest values being reached at $\lambda = 10^{-3}$ on the SpecTIR and Salinas datasets.

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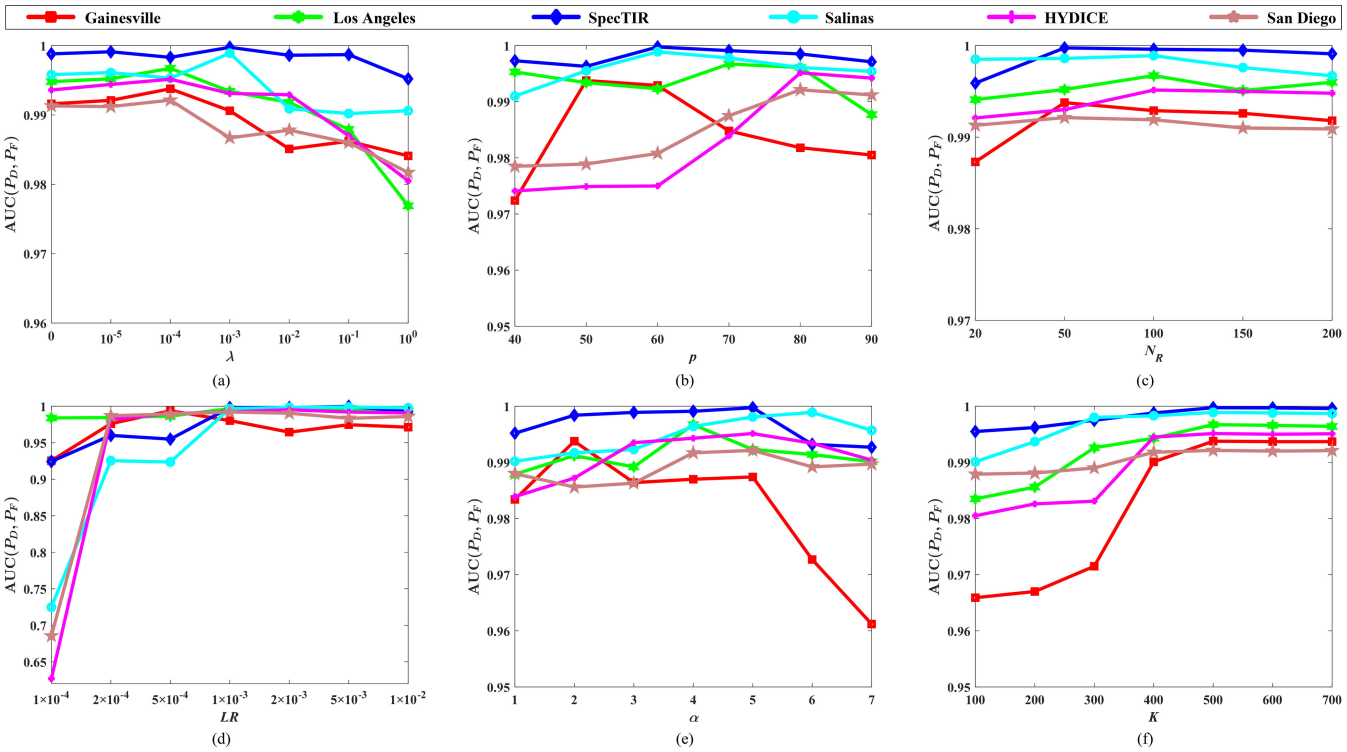


Fig. 4. Impacts of the different parameters for the proposed MAAE on the six datasets. (a) Impact of the tradeoff parameter λ. (b) Impact of the parameter p. (c) Impact of the number of superpixel regions. (d) Impact of LR. (e) Impact of the parameter α. (f) Impact of the number of memory items.

2) Parameter Analysis of p: The top p percent is quantitatively analyzed to verify its impact on the proposed MAAE. As shown in Fig. 4(b), the AUC(P_D, P_F) of our method obtains the highest values when $p = 50\%$ for the Gainesville dataset, $p = 70\%$ for the Los Angeles dataset, $p = 60\%$ for the SpecTIR and Salinas datasets, $p = 80\%$ for the HYDICE and San Diego datasets.

3) Parameter Analysis of N_R: The N_R (number of superpixel regions) is quantitatively analyzed on the six datasets. As shown in Fig. 4(c), the parameter N_R is configured as $\{20, 50, 100, 150, 200\}$. Observations show that the AUC(P_D, P_F) of our method obtains the highest values when $N_R = 50$ for the Gainesville, SpecTIR, and San Diego datasets, $N_R = 100$ for the Los Angeles, Salinas, and HYDICE datasets. It can be also observed that given large superpixel regions N_R, the AUC(P_D, P_F) values are relatively stable.

4) Parameter Analysis of LR: The LR is quantitatively analyzed on the six datasets. As shown in Fig. 4(d), the AUC(P_D, P_F) of our method obtains the highest values when LR $= 0.0005$ for the Gainesville dataset, LR $= 0.001$ for the San Diego and Los Angeles datasets, LR $= 0.005$ for the SpecTIR and Salinas datasets, and LR $= 0.002$ for the HYDICE dataset.

5) Parameter Analysis of α: The α is quantitatively analyzed to verify its impact on the proposed MAAE model. As shown in Fig. 4(e), the AUC(P_D, P_F) of our method achieves the highest values when $\alpha = 2$ for the Gainesville dataset, $\alpha = 4$ for Los Angeles dataset, $\alpha = 5$ for SpecTIR, HYDICE, and San Diego datasets, and $\alpha = 6$ for Salinas dataset.

6) Parameter Analysis of K: The K (the number of memory items) is quantitatively analyzed on the six datasets. As depicted in Fig. 4(f), the parameter K is configured as $\{100, 200, 300, 400, 500, 600, 700\}$. Observations show that the AUC(P_D, P_F) values exhibit a progressive increase when K varies from 100 to 500, with the highest values being reached at $K = 500$. Meanwhile, it can be also observed that given large memory items K, the AUC(P_D, P_F) values are relatively stable. Based on the above analysis, the K is set to 500 for the proposed MAAE in this article.

Conveniently, the parameter settings are listed in Table I.

In actual application, AD techniques can be used without any GT. Therefore, those are some possible insights for users to choose parameters without any GT.

1) *Parameter Choice Based on Prior Knowledge:* In some cases, we may have some prior knowledge about the data, such as the distribution characteristics of the data, the size of anomalies, and the possible number of anomalies, etc. This prior knowledge can serve as a basis for choosing parameters. For example, if superpixel segmentation is used in the AD method, a reasonable number of segmentation regions can be set according to the size or number of anomalies.
2) *Parameter Setting Based on Experience:* If the LR parameter is used in the AD method, a reasonable LR can be set based on experience, which can be adjusted by observing changes in loss values; if the memory module is employed in AD method, the model can usually achieve robust results when the memory size is set to a larger value.

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3) *Parameter Adjustment Based on Detection Results:* The parameters can be adjusted according to the detection results. For example, the visual detection map can directly qualitatively evaluate the model performance; using statistical methods (e.g., anomaly background separation map) to analyze the data distribution of detection results to qualitatively evaluate the separation degree between background and anomalies, or the background suppressibility.

These possible insights can better guide the adjustment of parameters, and alleviate the problem to a certain extent that the AD method may be sensitive to the choice of its parameters.

C. Ablation Study

To validate the influence of the proposed SAWC, SAM, and ESA, an ablation study with different configurations is compared with each other. As shown in Table II, the first line represents the MAE (i.e., baseline) method. The second line represents the baseline method combined with the SAWC, which is denoted as MAE + SAWC. The third line represents the baseline method combined with SAM, which is denoted as MAE + SAM. The fourth line represents the baseline method combined with ESA, which is denoted as MAE + ESA. The fifth line represents the baseline method combined with the SAWC and SAM, which is denoted as MAE + SAWC + SAM. The sixth line represents the baseline method combined with the SAWC and ESA, which is denoted as MAE + SAWC + ESA. The seventh line represents the baseline method combined with the SAM and ESA, which is denoted as MAE + SAM + ESA. The last line represents the proposed method, which is denoted as MAE + SAWC + SAM + ESA.

1) Influence of SAWC: To disclose the influence of the SAWC, we report the performance of our method with and without SAWC in Table II. Specifically, the AUC(P_D, P_F) values of our method with SAWC separately obtain the increase of 2.74%, 1.18%, 1.34%, 1.07%, 1.26%, and 2.54% compared with baseline on the Gainesville, Los Angeles, SpecTIR, Salinas, HYDICE, and San Diego datasets, which validates the effectiveness of SAWC.

2) Influence of SAM: To verify the influence of the SAM, the ablation study with and without SAM is displayed in Table II. Specifically, the AUC(P_D, P_F) values of our method with SAM separately obtain the increase of 2.86%, 2.31%, 1.44%, 2.12%, 2.13%, and 2.97% compared with baseline on the Gainesville, Los Angeles, SpecTIR, Salinas, HYDICE, and San Diego datasets, which validates the effectiveness of SAM.

3) Influence of ESA: To validate the influence of the ESA, we report the performance of our method with and without ESA in Table II. Specifically, the AUC(P_D, P_F) values of our method with ESA separately obtain the increase of 2.61%, 2.16%, 1.55%, 1.53%, 2.10%, and 2.90% compared with baseline on the Gainesville, Los Angeles, SpecTIR, Salinas, HYDICE, and San Diego datasets, which verifies the effectiveness of ESA.

4) Influence of Combination: The collaboration between SAWC and SAM yields varying degrees of improvement over using each individually, confirming their combined effectiveness. Similarly, the partnership between SAWC and ESA demonstrates enhanced performance compared to their individual use, illustrating the benefits of their integration. The combination of SAM and ESA shows different levels of improvement compared to using each individually, verifying their combined effectiveness. Meanwhile, the collaboration of SAWC, SAM, and ESA yields the highest AUC(P_D, P_F) compared to other configuration methods across the six datasets, illustrating the overall effectiveness of our method.

In addition, we also constructed another experiment where the best parameters were chosen for each variant to determine which module was the most important. As shown in Table III, the SAM module achieves the optimal AUC(P_D, P_F) values on the Gainesville and Los Angeles datasets, followed by SAWC and ESA modules. Therefore, the SAM is the most important module, next are the SAWC and ESA modules on the Gainesville and Los Angeles datasets.

D. Comparison With Other Advanced HAD Methods

To further validate the performance of the proposed MAAE, it is qualitatively and quantitatively compared with another 12 advanced HAD methods on six HSI datasets. The 12 HAD methods include RX [20], CRD [26], LRASR [30], LSDM-MOG [31], NJCR [63], RGAE [38], GAED [64], PCA-TLRSR [65], GNLTR [66], AHMID [67], MAENet [40], and OTCMA [68]. The parameters within all comparison methods are configured to achieve optimal performance.

1) Qualitative Comparison: Extensive qualitative comparison experiments, including ROC curve comparisons, anomaly background separation map comparisons, and detection map comparisons, are conducted to validate the superiority of the proposed MAAE.

a) ROC curve comparisons: Figs. 5–7 illustrate the 2-D (P_D, P_F) ROC curves, 2-D (P_D, τ) ROC curves, and 2-D (P_F, τ) ROC curves, respectively, for 13 HAD methods across six HSI datasets. Specifically, the 2-D (P_D, P_F) ROC curves for the 13 HAD methods across six HSI datasets are displayed as shown in Fig. 5(a)–(f). Observations show that the ROC curves of the proposed MAAE obtain higher P_D for the same P_F in most situations, indicating that the MAAE achieves superior overall detection performance compared to other HAD methods. The 2-D (P_D, τ) ROC curves for the 13 HAD methods across six HSI datasets are illustrated in Fig. 6(a)–(f). Observations show that the ROC curves of the MAAE obtain higher P_D for the same τ in most situations, which means the detection probability of the MAAE is competitive compared to other HAD methods. The 2-D (P_F, τ) ROC curves of the 13 HAD methods across six HSI datasets are demonstrated as shown in Fig. 7(a)–(f). Observations show that the ROC curves of the MAAE reside on the left side in most datasets, which means the false alarm rate of the MAAE is superior compared to most of the HAD methods. Meanwhile, the 3-D (P_D, P_F, τ) ROC curves and anomaly background separation maps are demonstrated in the Appendixes A and B, respectively.

b) Detection map comparisons: The detection maps of different HAD methods are subjectively compared with each

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TABLE II

ABLATION STUDY ON SIX HSI DATASETS OVER AUC(P_D, P_F) VALUES

MAE (Baseline)	SAWC	SAM	ESA	Gainesville	Los Angeles	SpecTIR	Salinas	HYDICE	San Diego
				0.9571	0.9678	0.9782	0.9696	0.9680	0.9606
	✓			0.9845	0.9796	0.9916	0.9803	0.9806	0.9860
		✓		0.9857	0.9909	0.9926	0.9908	0.9893	0.9903
			✓	0.9832	0.9894	0.9937	0.9849	0.9890	0.9896
✓	✓	✓		0.9916	0.9948	0.9988	0.9958	0.9936	0.9913
	✓		✓	0.9875	0.9924	0.9941	0.9914	0.9922	0.9911
		✓	✓	0.9898	0.9944	0.9988	0.9948	0.9917	0.9907
	✓	✓	✓	**0.9938**	**0.9967**	**0.9997**	**0.9989**	**0.9951**	**0.9921**

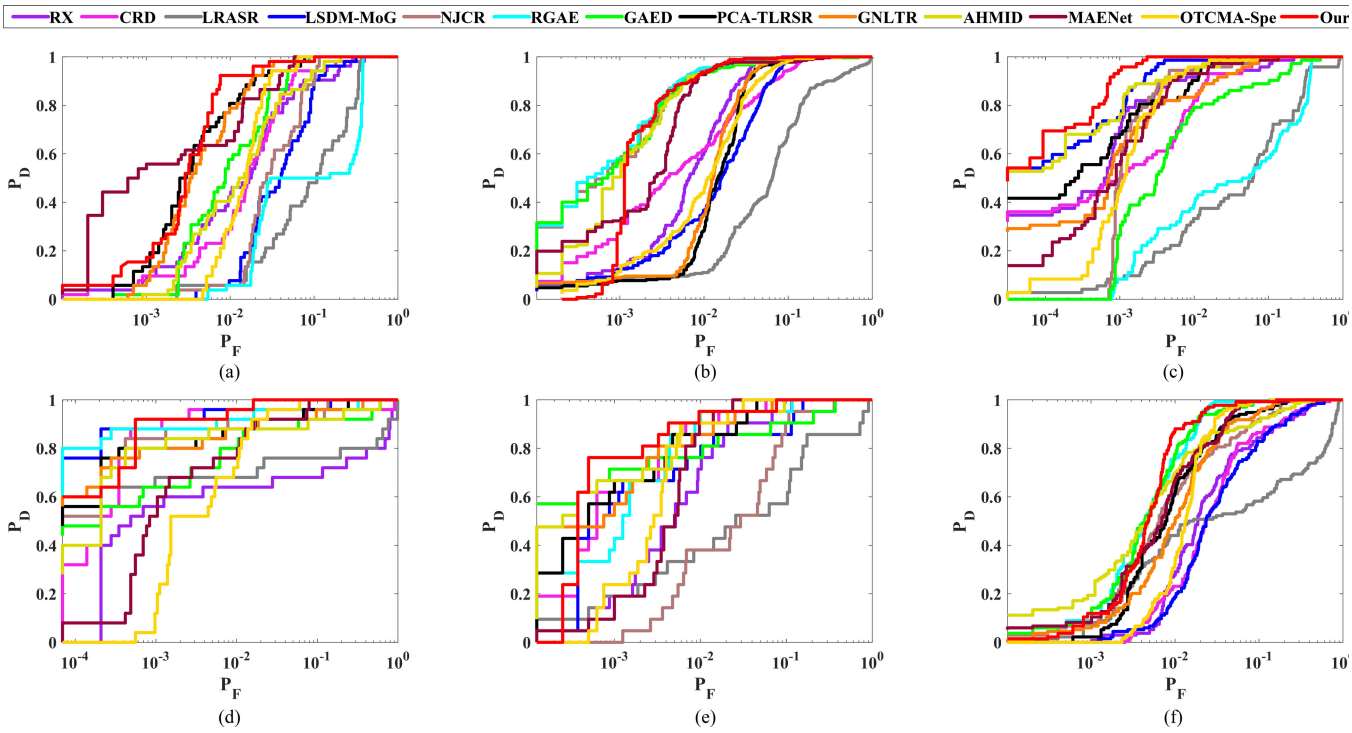


Fig. 5. Two-dimensional (P_D, P_F) ROC curves of different methods across six HSI datasets. (a) Gainesville. (b) Los Angeles. (c) SpecTIR. (d) Salinas. (e) HYDICE. (f) San Diego.

TABLE III

COMPARISON OF AUC(P_D, P_F) VALUES UNDER THE BEST PARAMETERS FOR EACH MODULE ON THE GAINESVILLE AND LOS ANGELES DATASETS

MAE (Baseline)	SAWC	SAM	ESA	Gainesville	Los Angeles
				0.9571	0.9678
	✓			0.9851	0.9919
✓		✓		**0.9888**	**0.9926**
			✓	0.9837	0.9905

other across six HSI datasets, as shown in Fig. 8. Observations show that the MAAE can capture anomalies to a large extent and generate fewer false alarms compared to most HAD methods on the six datasets. Specifically, for the Gainesville dataset, most of the HAD methods can capture as many anomalies as possible, except for the RGAE, which has terribly missed detection. The LRASR and OTCMA have higher false alarms. In addition, our method can better suppress the complex background relative to other HAD methods. For the Los Angeles dataset, most of the HAD methods capture as many anomalies as possible, except for the AHMID, which has terribly missed detection. Meanwhile, the MAAE can better distinguish anomalies from the background. For the SpecTIR dataset, most HAD methods are capable of locating anomalies, except for the LRASR, which generates more false alarm pixels. Furthermore, observations show that the MAAE demonstrates a superior ability to suppress complex backgrounds when compared to other HAD methods. For the Salinas dataset, the RX, LSDM-MOG, and GAED exhibit missed detection to some extent. The LRASR, NJCR, RGAE, PCA-TLRSR, GNLTR, and OTCMA have higher false alarms. The CRD, AHMID, and MAAE can detect all anomalies as much as possible, and the MAAE can better suppress background. For the HYDICE dataset, the LRASR, NJCR, RGAE, PCA-TLRSR, GNLTR, and OTCMA have higher false alarms. The rest HAD methods can locate as many anomalies as possible. Although GAED and MAAE both have a few missed detections to some extent, these two methods can better suppress background. For the San Diego dataset, the

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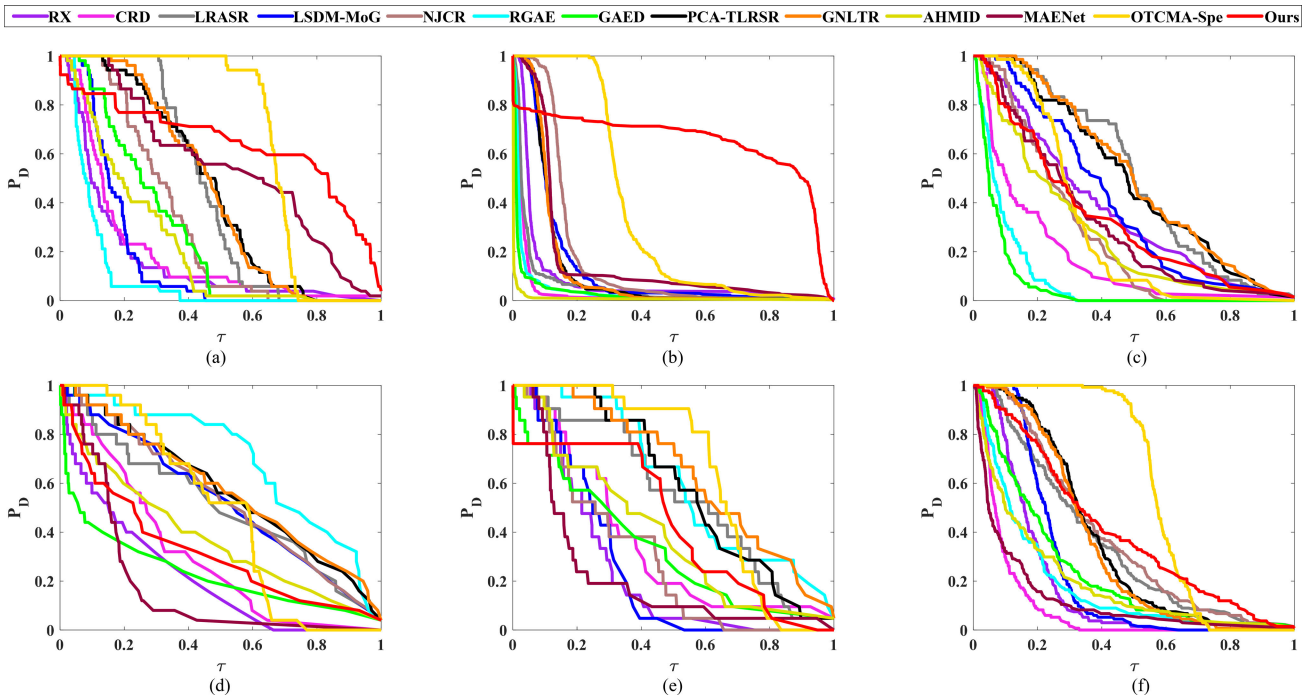


Fig. 6. Two-dimensional (P_D, τ) ROC curves of different methods across six HSI datasets. (a) Gainesville. (b) Los Angeles. (c) SpecTIR. (d) Salinas. (e) HYDICE. (f) San Diego.

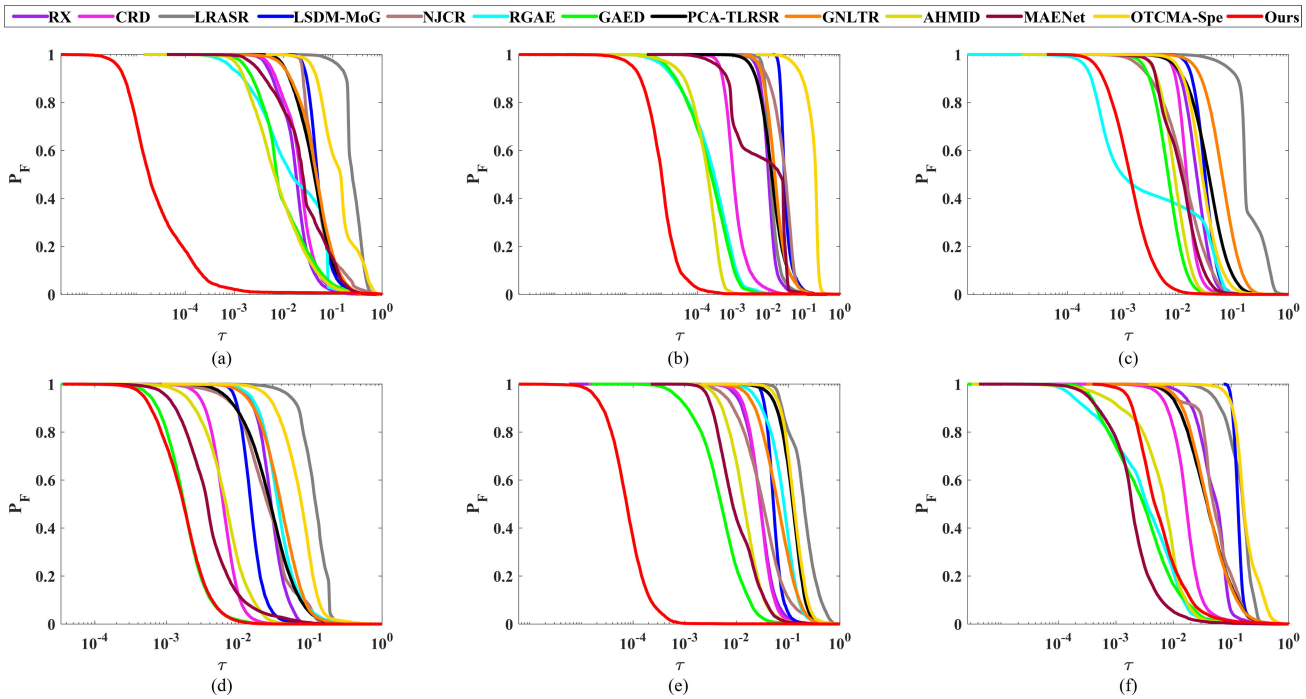


Fig. 7. Two-dimensional (P_F, τ) ROC curves of different methods across six HSI datasets. (a) Gainesville. (b) Los Angeles. (c) SpecTIR. (d) Salinas. (e) HYDICE. (f) San Diego.

RX, CRD, and MAENet cannot clearly display the shape of an airplane. The LRASR, LSDM-MOG, NJCR, PCA-TLRSR, GNLTR, and OTCMA have higher false alarms. The RGAE, GAED, AHMID, and MAAE achieve a good balance between detection accuracy and false alarms.

All in all, our method can locate as many anomalies as possible and better suppress complex backgrounds relative to most of the advanced HAD methods across six HSI datasets.

2) Quantitative Comparison:

a) Eight AUC metric comparisons: Extensive quantitative comparison experiments are further conducted to validate the detection performance of the MAAE. As shown in Table IV, eight different AUC metrics with respect to the 3-D ROC curve are reported for 13 HAD methods on the six datasets. The first- and second-best values are represented by bold and underlined numbers, respectively. Observations

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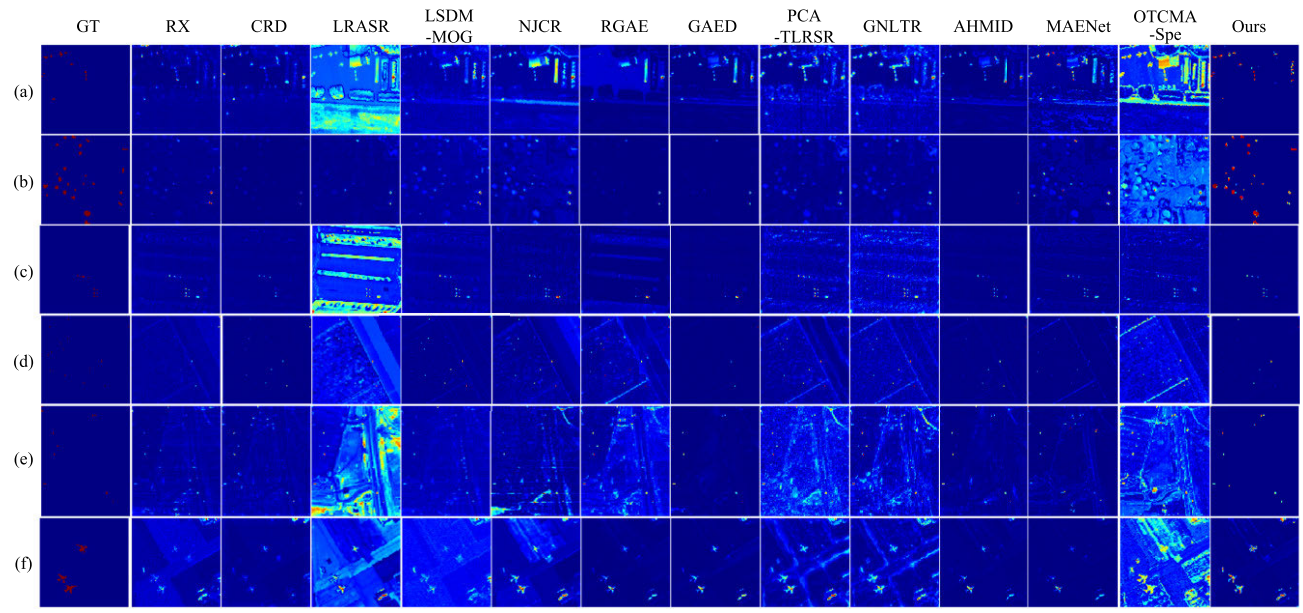


Fig. 8. Detection map comparisons with different HAD methods. (a) Gainesville. (b) Los Angeles. (c) SpecTIR. (d) Salinas. (e) HYDICE. (f) San Diego.

TABLE IV

AUC VALUES COMPARISON FOR DIFFERENT HAD METHODS ACROSS SIX HSI DATASETS

Dataset	AUC Values	RX [20]	CRD [26]	LRASR [30]	LSDM-MOG [31]	NJCR [63]	RGAE [38]	GAED [64]	PCA-TLRSR [65]	GNLTR [66]	AHMID [67]	MAENet [40]	OTCMA-Spe [68]	Ours
Gainesville	AUC(P_D, P_F)	0.9597	0.9697	0.8559	0.9438	0.9580	0.8177	0.9833	0.9912	0.9923	0.9721	0.9901	0.9823	**0.9938**
	AUC(P_D, τ)	0.1665	0.1940	0.4438	0.1742	0.3190	0.1010	0.2760	0.4491	0.4495	0.2244	0.5507	**0.6724**	0.6408
	AUC(P_F, τ)	0.0229	0.0287	0.2833	0.0619	0.0722	0.0382	0.0229	0.0581	0.0638	0.0200	0.0474	0.1830	**0.0048**
	AUC_{TD}	1.1262	1.1636	1.2997	1.1180	1.2770	0.9187	1.2594	1.4403	1.4418	1.1965	1.5408	**1.6547**	1.6346
	AUC_{BS}	0.9367	0.9410	0.5726	0.8819	0.8858	0.7795	0.9604	0.9331	0.9284	0.9521	0.9427	0.7993	**0.9890**
	AUC_{ODP}	1.1033	1.1349	1.0164	1.0561	1.2048	0.8805	1.2365	1.3822	1.3779	1.1765	1.4934	1.4717	**1.6298**
	AUC_{TDBS}	0.1436	0.1653	0.1604	0.1123	0.2468	0.0628	0.2531	0.3910	0.3857	0.2044	0.5033	0.4894	**0.6360**
	AUC_{SNPR}	7.2663	6.7610	1.5663	2.8132	4.4189	2.6441	12.0465	7.7307	7.0405	11.2314	11.6271	3.6739	**134.5210**
Los Angeles	AUC(P_D, P_F)	0.9887	0.9794	0.8748	0.9745	0.9961	0.9948	0.9927	0.9805	0.9833	0.9943	0.9924	0.9809	**0.9967**
	AUC(P_D, τ)	0.0891	0.0366	0.0590	0.1362	0.1754	0.0393	0.0321	0.1230	0.1234	0.0103	0.1637	0.3652	**0.6470**
	AUC(P_F, τ)	0.0114	0.0018	0.0144	0.0331	0.0311	0.0006	0.0005	0.0185	0.0204	**0.0002**	0.0165	0.1903	0.0019
	AUC_{TD}	1.0778	1.0161	0.9339	1.1107	1.1716	1.0342	1.0249	1.1036	1.1067	1.0046	1.1561	1.3461	**1.6437**
	AUC_{BS}	0.9773	0.9777	0.8604	0.9414	0.9651	0.9942	0.9922	0.9620	0.9629	0.9941	0.9759	0.7906	**0.9949**
	AUC_{ODP}	1.0664	1.0143	0.9194	1.0776	1.1405	1.0335	1.0244	1.0851	1.0863	1.0044	1.1395	1.1558	**1.6419**
	AUC_{TDBS}	0.0777	0.0349	0.0446	0.1031	0.1444	0.0387	0.0317	0.1045	0.1030	0.0100	0.1472	0.1749	**0.6451**
	AUC_{SNPR}	7.8088	20.8810	4.0911	4.1152	5.6476	65.3356	66.5770	6.6497	6.0484	43.9356	9.9064	1.9189	**349.1495**
SpecTIR	AUC(P_D, P_F)	0.9914	0.9920	0.8714	0.9991	0.9965	0.8777	0.9703	0.9972	0.9931	0.9976	0.9956	0.9971	**0.9997**
	AUC(P_D, τ)	0.3683	0.1743	0.5095	0.4003	0.2675	0.0889	0.0690	0.4993	**0.5144**	0.2746	0.3150	0.3063	0.3348
	AUC(P_F, τ)	0.0249	0.0159	0.2271	0.0337	0.0187	0.0176	0.0084	0.0455	0.0749	0.0101	0.0149	0.0346	**0.0021**
	AUC_{TD}	1.3597	1.1662	1.3809	1.3993	1.2640	0.9666	1.0393	1.4965	**1.5075**	1.2722	1.3105	1.3034	1.3346
	AUC_{BS}	0.9665	0.9761	0.6444	0.9654	0.9778	0.8600	0.9619	0.9517	0.9182	0.9875	0.9806	0.9625	**0.9976**
	AUC_{ODP}	1.3348	1.1503	1.1538	1.3657	1.2453	0.9490	1.0310	**1.4510**	1.4326	1.2621	1.2956	1.2688	1.3324
	AUC_{TDBS}	0.3434	0.1584	0.2824	0.3666	0.2488	0.0713	0.0606	**0.4538**	0.4395	0.2645	0.3000	0.2717	0.3327
	AUC_{SNPR}	14.8183	10.9570	2.2436	11.8848	14.3415	5.0408	8.2345	10.9746	6.8685	27.1178	21.0813	8.8622	**156.3146**
Salinas	AUC(P_D, P_F)	0.8073	0.9635	0.8332	0.9939	0.9892	0.9862	0.9528	0.9864	0.9802	0.9643	0.9911	0.9923	**0.9989**
	AUC(P_D, τ)	0.2143	0.3012	0.5058	0.5273	0.5382	**0.6949**	0.2318	0.5706	0.5769	0.3750	0.1764	0.4795	0.3219
	AUC(P_F, τ)	0.0314	0.0069	0.1227	0.0166	0.0347	0.0447	**0.0023**	0.0366	0.0467	0.0083	0.0070	0.0829	**0.0023**
	AUC_{TD}	1.0216	1.2647	1.3390	1.5213	1.5275	**1.6811**	1.1846	1.5570	1.5571	1.3392	1.1675	1.4719	1.3208
	AUC_{BS}	0.7759	0.9566	0.7106	0.9773	0.9546	0.9416	0.9505	0.9498	0.9335	0.9559	0.9841	0.9094	**0.9966**
	AUC_{ODP}	0.9903	1.2577	1.2163	1.5047	1.4928	**1.6364**	1.1823	1.5203	1.5104	1.3309	1.1605	1.3889	1.3185
	AUC_{TDBS}	0.1829	0.2942	0.3831	0.5107	0.5036	**0.6502**	0.2295	0.5339	0.5302	0.3666	0.1694	0.3966	0.3196
	AUC_{SNPR}	6.8318	43.4025	4.1237	31.7153	15.5241	15.5514	100.1417	15.5682	12.3493	44.9832	25.2389	5.7816	**141.3768**
HYDICE	AUC(P_D, P_F)	0.9857	**0.9951**	0.8402	0.9792	0.9470	0.9923	0.9679	0.9940	0.9939	0.9938	0.9943	**0.9951**	**0.9951**
	AUC(P_D, τ)	0.2404	0.3545	0.5519	0.2502	0.2927	0.6072	0.3337	0.5903	0.6454	0.3933	0.2187	**0.6466**	0.4371
	AUC(P_F, τ)	0.0351	0.0333	0.2319	0.0570	0.0532	0.0932	0.0081	0.1326	0.0822	0.0188	0.0164	0.1465	**0.0006**
	AUC_{TD}	1.2261	1.3495	1.3921	1.2294	1.2397	1.5995	1.3016	1.5842	1.6393	1.3871	1.2130	**1.6416**	1.4322
	AUC_{BS}	0.9506	0.9618	0.6083	0.9222	0.8938	0.8991	0.9598	0.8614	0.9117	0.9750	0.9779	0.8485	**0.9946**
	AUC_{ODP}	1.1910	1.3163	1.1602	1.1724	1.1865	1.5062	1.2935	1.4516	**1.5571**	1.3683	1.1966	1.4951	1.4316
	AUC_{TDBS}	0.2053	0.3212	0.3200	0.1932	0.2395	0.5139	0.3256	0.4577	**0.5632**	0.3745	0.2023	0.5000	0.4365
	AUC_{SNPR}	6.8442	10.6551	2.3799	4.3907	5.5027	6.5126	41.2122	4.4519	7.8548	20.9279	13.3494	4.4126	**750.2439**
San Diego	AUC(P_D, P_F)	0.9403	0.9412	0.7696	0.9339	0.9736	0.9919	0.9907	0.9800	0.9764	0.9732	0.9857	0.9835	**0.9921**
	AUC(P_D, τ)	0.1778	0.0911	0.3471	0.2381	0.3807	0.1807	0.2337	0.3584	0.3425	0.1930	0.1225	**0.5828**	0.4030
	AUC(P_F, τ)	0.0589	0.0214	0.1599	0.1365	0.0692	0.0075	0.0079	0.0589	0.0592	0.0094	**0.0040**	0.2046	0.0129
	AUC_{TD}	1.1181	1.0323	1.1167	1.1720	1.3542	1.1727	1.2244	1.3385	1.3189	1.1662	1.1082	**1.5663**	1.3951
	AUC_{BS}	0.8814	0.9199	0.6097	0.7974	0.9044	**0.9845**	0.9828	0.9211	0.9172	0.9639	0.9817	0.7788	0.9792
	AUC_{ODP}	1.0592	1.0110	0.9568	1.0355	1.2850	1.1652	1.2165	1.2796	1.2597	1.1569	1.1042	1.3616	**1.3822**
	AUC_{TDBS}	0.1189	0.0698	0.1872	0.1016	0.3114	0.1733	0.2258	0.2996	0.2832	0.1837	0.1185	0.3782	**0.3900**
	AUC_{SNPR}	3.0176	4.2658	2.1703	1.7441	5.5001	24.2040	29.5755	6.0873	5.7840	20.6407	30.5280	2.8481	**31.1509**

show that the MAAE achieves the highest values in terms of AUC(P_D, P_F) and AUC_{SNPR} on six datasets. For the AUC(P_D, τ) metric, the OTCMA obtains the best detection performance on the Gainesville, HYDICE, and San Diego

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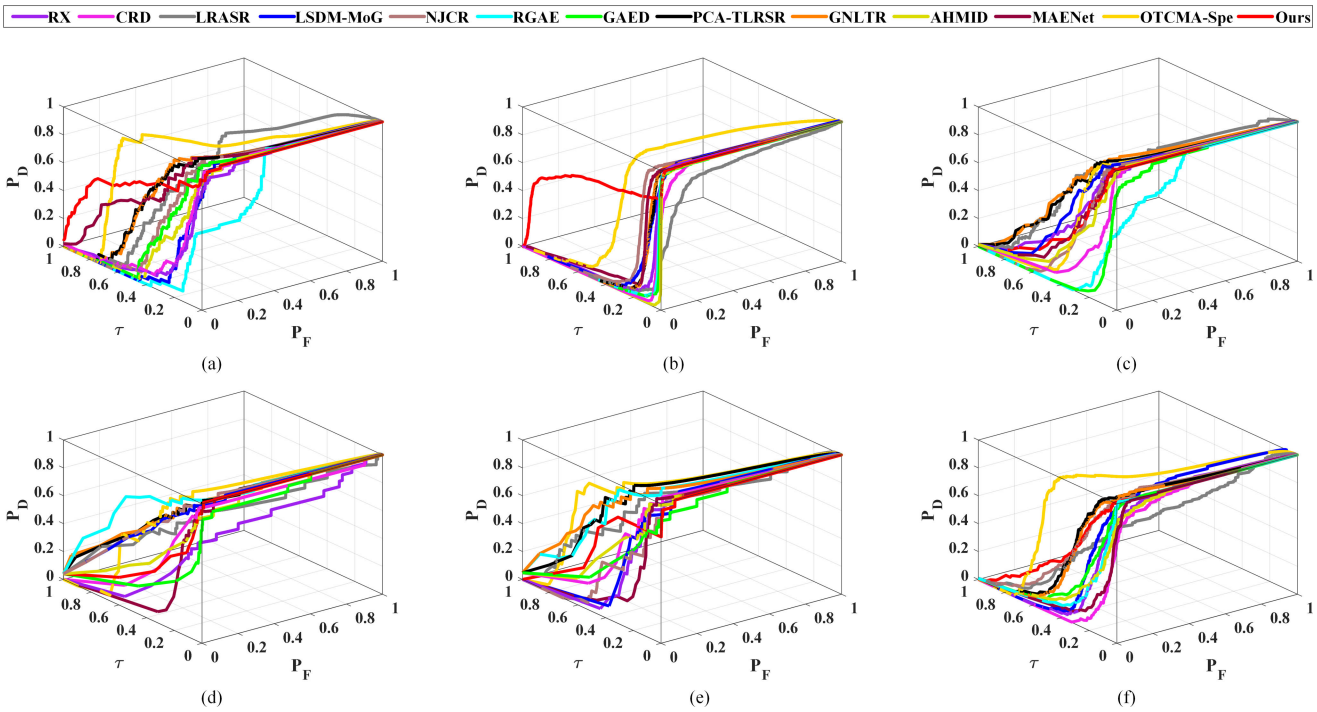


Fig. 9. Three-dimensional (P_D, P_F, τ) ROC curves of different methods across six HSI datasets. (a) Gainesville. (b) Los Angeles. (c) SpecTIR. (d) Salinas. (e) HYDICE. (f) San Diego.

TABLE V
TRAINING TIME (IN SECOND) COMPARISON FOR DIFFERENT HAD METHODS ACROSS SIX HSI DATASETS

Datasets	RGAE [38]	GAED [64]	MAENet [40]	OTCMA-Spe [68]	Ours-GPU / CPU
Gainesville	78.7731	**43.4580**	120.1197	416.5146	162.6150 / 254.0755
Los Angeles	85.2036	**44.5882**	124.3765	518.2044	255.4378 / 353.3656
SpecTIR	209.2537	**135.3291**	329.3242	1243.3930	495.0160 / 1108.7416
Salinas	124.7273	**66.0866**	170.7285	734.3465	264.7288 / 472.8081
HYDICE	57.3693	**35.1004**	95.5381	326.5860	151.1266 / 219.7472
San Diego	77.7482	**43.0067**	123.2166	418.7419	183.0602 / 276.1631
Average	105.5125	**61.2615**	160.5506	609.6311	251.9974 / 447.4835

datasets, followed by RGAE, GNLTR, and MAAE, which obtain the best scores across the Salinas, SpecTIR, and Los Angeles datasets, respectively. For the AUC(P_F, τ) metric, the MAAE obtains the lowest false alarm rates on the Gainesville, SpecTIR, Salinas, and HYDICE datasets. The AHMID, GAED, and MAENet obtained the lowest false alarm rate on the Los Angeles, Salinas, and San Diego datasets, respectively. For the AUC_{TD} metric, the OTCMA obtains the optimal values across the Gainesville, HYDICE, and San Diego datasets. The RGAE, GNLTR, and MAAE obtain the highest values on the Salinas, SpecTIR, and Los Angeles datasets, respectively. For the AUC_{BS} metric, the MAAE obtains the best background suppressibility across all comparison datasets, with the sole exception of the San Diego dataset, on which RGAE yields the highest value. For the AUC_{ODP} and AUC_{TDBS} metrics, the MAAE achieves the optimal values across the Gainesville, Los Angeles, and San Diego datasets. The PCA-TLRSR, RGAE, and GNLTR obtain the highest values on the SpecTIR, Salinas, and HYDICE datasets, respectively.

In a word, extensive quantitative comparison experiments effectively demonstrate that the MAAE achieves superior and reliable detection performance relative to most of the advanced HAD methods across six HSI datasets.

b) Running time comparisons: To evaluate the efficiency of the MAAE, the training and inference time for different HAD methods across six HSI datasets are reported in Tables V and VI, respectively. Specifically, the RX method in statistic-based HAD methods does not require model optimization. Therefore, only the inference time for the RX method is reported in Table VI. The representation-based HAD methods belong to the iterative recursive optimization algorithms. These methods need to be optimized from scratch each time, hence they do not have a so-called training stage, only an inference stage. Therefore, following reference [69], only the inference time for these methods is reported in Table VI. The reconstruction-based HAD methods can be divided into two stages (i.e., training and inference), and these two stages can be executed separately. Therefore, following reference [69], both the training and inference time for these methods are reported in Tables V and VI, respectively.

As shown in Table V, the GAED takes the shortest training time, and the OTCMA-Spe consumes the longest training time across six datasets. Our method has a longer training time compared to the RGAE, GAED, and MAENet, and most of the time is spent on the SAWC module, as it involves the calculation of AWs for each sample. Meanwhile, we also report the training time that is implemented on a CPU-only machine. It can be seen that although our method is implemented on the CPU, it is still faster than OTCMA-Spe that implemented on the GPU.

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TABLE VI

INFERENCE TIME (IN SECOND) COMPARISON FOR DIFFERENT HAD METHODS ACROSS SIX HSI DATASETS

Datasets	RX [20]	CRD [26]	LRASR [30]	LSDM-MOG [31]	NJCR [63]	RGAE [38]	GAED [64]	PCA-TLRSR [65]	GNLTR [66]	AHMID [67]	MAENet [40]	OTCMA-Spe [68]	Ours
Gainesville	**0.0574**	2.5544	16.7953	7.2765	6.1318	0.1430	0.1506	8.1414	2.7577	38.2966	1.8060	3.1999	1.8646
Los Angeles	**0.0549**	3.5248	12.0713	11.2326	9.4755	0.1324	0.1514	8.4914	2.7203	44.9557	1.8510	3.5029	1.8561
SpecTIR	**0.0944**	8.2566	37.3347	31.8114	23.9106	0.1639	0.2273	28.5509	12.6895	74.7583	1.9397	5.6866	2.1601
Salinas	**0.0736**	4.7403	16.0340	9.1539	11.4132	0.1447	0.1713	11.6770	6.1689	44.1865	1.8680	3.7463	1.9471
HYDICE	**0.0739**	2.5012	31.7948	7.3256	6.0493	0.1264	0.1462	6.3367	3.5174	21.3605	1.7856	2.9670	1.8441
San Diego	**0.0508**	3.2478	14.9998	5.3204	7.7072	0.1442	0.1481	8.7012	2.7814	28.2716	1.7889	3.2437	1.8778
Average	**0.0675**	4.1375	21.5050	12.0201	10.7813	0.1424	0.1658	11.9831	5.1059	41.9715	1.8399	3.7244	1.9250

TABLE VII

HARDWARE RESOURCE LOAD OF DIFFERENT HAD ALGORITHMS DURING INFERENCE ON THE GAINESVILLE DATASET

Resources	RX [20]	CRD [26]	LRASR [30]	LSDM-MOG [31]	NJCR [63]	RGAE [38]	GAED [64]	PCA-TLRSR [65]	GNLTR [66]	AHMID [67]	MAENet [40]	OTCMA-Spe [68]	Ours
CPU-util (%)	9.5	90.5	86.5	91.2	74.5	21.2	25.1	82.6	68.0	79.2	28.0	35.5	31.2
Memory (MB)	385.1	463.1	579.4	628.7	640.2	425.4	470.2	779.4	792.2	899.1	1521.1	4415.3	1520.1
GPU-util (%)	-	-	-	-	-	-	-	-	-	-	7.0	80.0	7.0
GPU Memory (MiB)	-	-	-	-	-	-	-	-	-	-	606	4006	604

TABLE VIII

RESOURCES AND TIME COMPARISONS BETWEEN OUR METHOD AND THESE HAD METHODS THAT PROVIDE SOURCE CODES AND SUPPORT PARALLELIZED RUNNING ON GPU MACHINES. THESE EXPERIMENTS ARE IMPLEMENTED ON THE GAINESVILLE DATASET, UBUNTU 16.04, AND TWO NVIDIA TITAN RTX GPUS

Resources & Time	Training phase			Inference phase		
	MAENet [40]	OTCMA-Spe [68]	Ours	MAENet [40]	OTCMA-Spe [68]	Ours
Mean GPU-util (%)	18.0	89.0	15.0	4.0	71.0	4.0
Mean GPU memory (MiB)	1160	7469	1515	644	4266	642
Time (s)	73.4335	250.7289	97.1669	1.5124	2.5170	1.5285

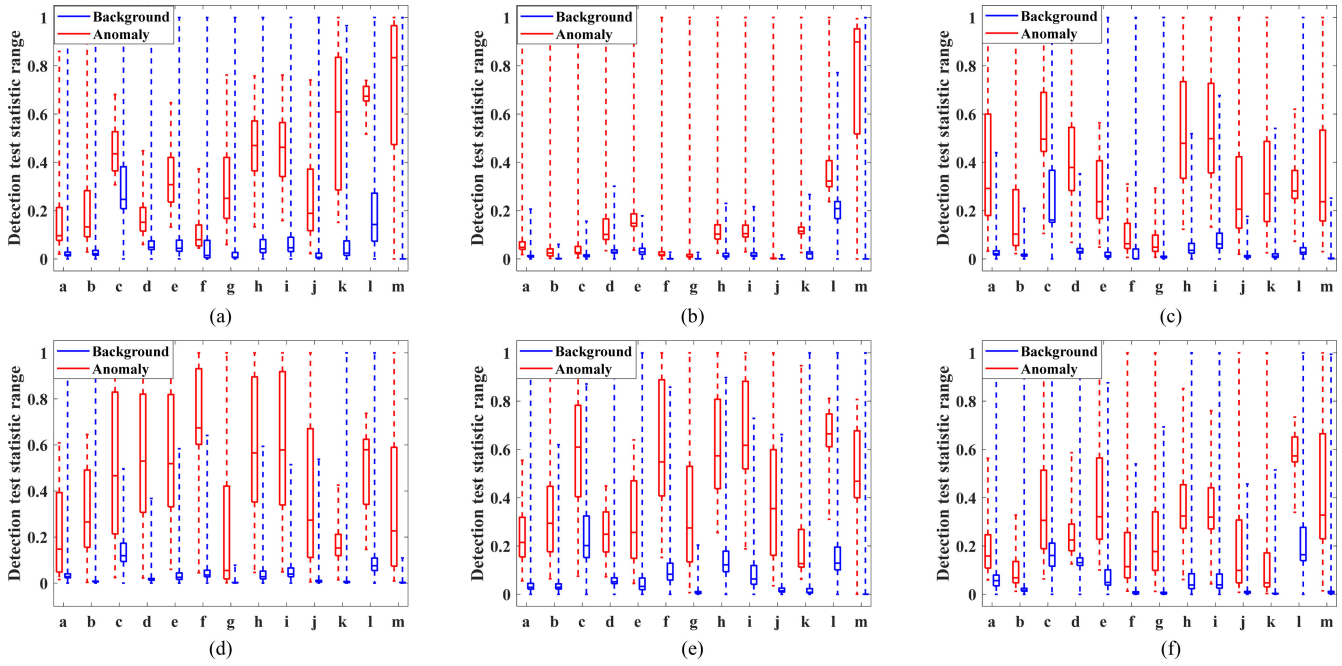


Fig. 10. Anomaly background separation maps of different methods across six HSI datasets. (a) Gainesville. (b) Los Angeles. (c) SpecTIR. (d) Salinas. (e) HYDICE. (f) San Diego. The a–m denote the RX, CRD, LRASR, LSDM-MOG, NJCR, RGAE, GAED, PCA-TLRSR, GNLTR, AHMID, MAENet, OTCMA-Spe, and MAAE, respectively.

As shown in Table VI, the RX takes the shortest inference time, and the AHMID consumes the longest inference time across six datasets. Our method has a longer inference time compared to the AE-based method, and most of the time is spent on the memory module, as it involves the retrieval of relevant memory items. Specifically, the average inference time of the MAAE is solely longer than that of RX, RGAE, and GAED, and nearly the same time as MAENet, which demonstrates that the efficiency of our method is competitive compared to most of the HAD methods.

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c) Resources and time comparisons: The hardware resource load of different HAD algorithms during inference on the Gainesville dataset are reported in Table VII. Specifically, as shown in the first and second lines, the RX has the lowest hardware resource load due to its simple network during the inference phase, while the OTCMA-Spe has the highest load because of its complex structure. For the CPU utilization, our method is competitive compared to most of the HAD methods. Meanwhile, these HAD algorithms, which support inference on GPU machines, are compared with our method. As shown in the third and fourth lines of Table VII, our method has a lower load than the OTCMA-Spe method and has nearly the same load as MAENet because of their similar structure.

In addition, our method is compared with two HAD methods that provide source codes and support parallelized running on GPU machines in terms of resources and time on the Gainesville dataset. Specifically, as shown in Table VIII, our method has a lower load than the OTCMA-Spe method and has nearly the same load as MAENet during training and inference phases. Meanwhile, when using two parallel GPUs for training and inference, both training and inference times are reduced to a certain extent compared to using a single GPU. Among them, the MAENet required the least amount of time, and OTCMA-Spe required the most time.

V. Conclusion

In this article, an MAAE model is devised to better reconstruct the background and suppress anomaly reconstruction for HAD in HSIs. Specifically, an SAWC module is developed to generate AWs that consider the contextual information in the error map. These AWs are then incorporated into the reconstruction loss, focusing the training on BSs by assigning them larger weights than anomalies. Consequently, the MAAE is beneficial for reconstructing BSs but is not conducive to abnormal ones. To suppress the interference of abnormal samples in the memory module, a SAM module is devised to explore sample attribution, and solely the mined BSs are employed to train the memory module. In addition, an ESA module is designed to further weaken the reconstruction ability of ASs. Consequently, collaborating the SAM module with the ESA module can effectively enhance the separation degree between anomalies and background, thereby improving the detection performance. The ablation study validated the effectiveness of the proposed SAWC, SAM, and ESA. Extensive comparison experiments on six HSI datasets demonstrated the superiority in terms of comprehensive detection performance and background suppression of our method. However, because of the design of SAWC and memory modules, the proposed method requires more training and inference time compared to traditional AE-based HAD methods. Therefore, exploring a lightweight HAD model is our future work.

Appendix A
Three-Dimensional (P_D, P_F, τ) ROC Curves Comparisons

See Fig. 9.

Appendix B
Anomaly Background Separation Map Comparisons

As shown in Fig. 10, the red and blue boxes represent the data distributions of anomalies and background, respectively. Specifically, for the Gainesville, Los Angeles, and HYDICE datasets, the separation degree of our method significantly outperforms that of other HAD methods, as shown in Fig. 10(a)–(e). For the SpecTIR dataset, the LSDM-MOG, PCA-TLRSR, and GNLTR acquire the largest separation distance, and the separation degree of MAENet, OTCMA-Spe, and MAAE follows closely, as shown in Fig. 10(c). For the Salinas dataset, although the MAAE did not achieve superior separation performance, it exhibits superior background suppressibility compared to other HAD methods, as illustrated in Fig. 10(d). For the San Diego dataset, the PCA-TLRSR, GNLTR, and MAAE achieve similar optimal separation degrees, as shown in Fig. 10(f).

In summary, our method exhibits superior separation performance and background suppression capability relative to most of the advanced HAD methods across six HSI datasets.

References

[1] Q. Wang, J. Lin, and Y. Yuan, "Salient band selection for hyperspectral image classification via manifold ranking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 6, pp. 1279–1289, Jun. 2016.

[2] N. He, L. Fang, S. Li, J. Plaza, and A. Plaza, "Skip-connected covariance network for remote sensing scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 5, pp. 1461–1474, May 2020.

[3] M. E. Paoletti, S. Moreno-Álvarez, Y. Xue, J. M. Haut, and A. Plaza, "AAtt-CNN: Automatic attention-based convolutional neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5511118.

[4] R. M. Willett, M. F. Duarte, M. A. Davenport, and R. G. Baraniuk, "Sparsity and structure in hyperspectral imaging: Sensing, reconstruction, and target detection," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 116–126, Jan. 2014.

[5] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.

[6] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 64–90, Mar. 2022.

[7] D. W. J. Stein, S. G. Beaven, L. E. Hoff, E. M. Winter, A. P. Schaum, and A. D. Stocker, "Anomaly detection from hyperspectral imagery," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 58–69, Jan. 2002.

[8] Chein-I. Chang, "Hyperspectral anomaly detection: A dual theory of hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511720.

[9] S. Khazai, S. Homayouni, A. Safari, and B. Mojaradi, "Anomaly detection in hyperspectral images based on an adaptive support vector method," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 4, pp. 646–650, Jul. 2011.

[10] M. Shimoni, R. Haelterman, and C. Perneel, "Hypersectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.

[11] S. Lin, M. Zhang, X. Cheng, K. Zhou, S. Zhao, and H. Wang, "Hyperspectral anomaly detection via sparse representation and collaborative representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 946–961, 2023.

[12] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2012, pp. 1097–1105.

[13] X. Cheng, M. Zhang, S. Lin, Y. Li, and H. Wang, "Deep self-representation learning framework for hyperspectral anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–16, 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:08:41 UTC from IEEE Xplore. Restrictions apply.

[14] X. Fu, S. Jia, L. Zhuang, M. Xu, J. Zhou, and Q. Li, "Hyperspectral anomaly detection via deep plug-and-play denoising CNN regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9553–9568, Nov. 2021.

[15] S. Lin, M. Zhang, X. Cheng, L. Shi, P. Gamba, and H. Wang, "Dynamic low-rank and sparse priors constrained deep autoencoders for hyperspectral anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–18, 2024.

[16] Y. Huo, X. Qian, C. Li, and W. Wang, "Multiple instance complementary detection and difficulty evaluation for weakly supervised object detection in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[17] D. Gong et al., "Memorizing normality to detect anomaly: Memory-augmented deep autoencoder for unsupervised anomaly detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1705–1714.

[18] Y. Xu, L. Zhang, B. Du, and L. Zhang, "Hyperspectral anomaly detection based on machine learning: An overview," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3351–3364, 2022.

[19] X. Hu et al., "Hyperspectral anomaly detection using deep learning: A review," *Remote Sens.*, vol. 14, no. 9, p. 1973, Apr. 2022.

[20] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Aug. 1990.

[21] H. Kwon, S. Z. Der, and N. M. Nasrabadi, "Adaptive anomaly detection using subspace separation for hyperspectral imagery," *Opt. Eng.*, vol. 42, no. 11, pp. 3342–3351, 2003.

[22] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RXD and linear filter-based RXD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.

[23] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[24] J. Zhou, C. Kwan, B. Ayhan, and M. T. Eismann, "A novel cluster kernel RX algorithm for anomaly and change detection using hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 11, pp. 6497–6504, Nov. 2016.

[25] A. Schaum, "Joint subspace detection of hyperspectral targets," in *Proc. IEEE Aerosp. Conf.*, vol. 3, Jul. 2004, p. 1824.

[26] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[27] M. Vafadar and H. Ghassemian, "Anomaly detection of hyperspectral imagery using modified collaborative representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 4, pp. 577–581, Apr. 2018.

[28] J. Li, H. Zhang, L. Zhang, and L. Ma, "Hyperspectral anomaly detection by the use of background joint sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2523–2533, Jun. 2015.

[29] Z. Yuan, H. Sun, K. Ji, Z. Li, and H. Zou, "Local sparsity divergence for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 10, pp. 1697–1701, Oct. 2014.

[30] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[31] L. Li, W. Li, Q. Du, and R. Tao, "Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4363–4372, Sep. 2020.

[32] Q. Xiao, L. Zhao, and S. Chen, "Tensor low-rank sparse representation learning for hyperspectral anomaly detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2023, pp. 7356–7359.

[33] S. Lin, M. Zhang, X. Cheng, K. Zhou, S. Zhao, and H. Wang, "Dual collaborative constraints regularized low-rank and sparse representation via robust dictionaries construction for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 2009–2024, 2023.

[34] Y.-P. Zhao, H. Li, Y. Chen, Z. Wang, and X. Li, "Hyperspectral anomaly detection via structured sparsity plus enhanced low-rankness," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515115.

[35] C. Li, B. Zhang, D. Hong, J. Yao, and J. Chanussot, "LRR-Net: An interpretable deep unfolding network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513412.

[36] E. Bati, A. Calíškan, A. Koz, and A. A. Alatan, "Hyperspectral anomaly detection method based on auto-encoder," *Proc. SPIE*, vol. 9643, pp. 220–226, Oct. 2015.

[37] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503314.

[38] G. Fan, Y. Ma, X. Mei, F. Fan, J. Huang, and J. Ma, "Hyperspectral anomaly detection with robust graph autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511314.

[39] Y. Liu, W. Xie, Y. Li, Z. Li, and Q. Du, "Dual-frequency autoencoder for anomaly detection in transformed hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5523613.

[40] Z. Zhao and B. Sun, "Hyperspectral anomaly detection via memory-augmented autoencoders," *CAAI Trans. Intell. Technol.*, vol. 8, no. 4, pp. 1274–1287, Dec. 2023.

[41] T. Jiang, W. Xie, Y. Li, J. Lei, and Q. Du, "Weakly supervised discriminative learning with spectral constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6504–6517, Nov. 2022.

[42] Y. Li, T. Jiang, W. Xie, J. Lei, and Q. Du, "Sparse coding-inspired GAN for hyperspectral anomaly detection in weakly supervised learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512811.

[43] K. Jiang, W. Xie, Y. Li, J. Lei, G. He, and Q. Du, "Semisupervised spectral learning with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 5224–5236, Jul. 2020.

[44] W. Xie, B. Liu, Y. Li, J. Lei, and Q. Du, "Autoencoder and adversarial-learning-based semisupervised background estimation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5416–5427, Aug. 2020.

[45] W. Xie, B. Liu, Y. Li, J. Lei, C.-I. Chang, and G. He, "Spectral adversarial feature learning for anomaly detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2352–2365, Apr. 2020.

[46] M. Díaz, R. Guerra, P. Horstrand, S. López, and R. Sarmiento, "A line-by-line fast anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8968–8982, Nov. 2019.

[47] L. Zhang and B. Cheng, "Transferred CNN based on tensor for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 12, pp. 2115–2119, Dec. 2020.

[48] X. Cheng, M. Zhang, S. Lin, K. Zhou, S. Zhao, and H. Wang, "Two-stream isolation forest based on deep features for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[49] M. Coca, I. C. Neagoe, and M. Datcu, "Hybrid DNN-Dirichlet anomaly detection and ranking: Case of burned areas discovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4414116.

[50] Z. Li, Y. Wang, C. Xiao, Q. Ling, Z. Lin, and W. An, "You only train once: Learning a general anomaly enhancement network with random masks for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506718.

[51] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

[52] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. A. Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.

[53] W. Xie, X. Zhang, Y. Li, J. Lei, J. Li, and Q. Du, "Weakly supervised low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 51, no. 8, pp. 3889–3900, Aug. 2021.

[54] J. A. Herweg et al., "SpecTIR hyperspectral airborne Rochester experiment data collection campaign," *Proc. SPIE*, vol. 8390, Apr. 2012, Art. no. 839028.

[55] R. Feng, H. Li, L. Wang, Y. Zhong, L. Zhang, and T. Zeng, "Local spatial constraint and total variation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512216.

[56] S. M. Schweizer and J. M. F. Moura, "Efficient detection in hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 10, no. 4, pp. 584–597, Apr. 2001.

[57] T. Jiang, Y. Li, W. Xie, and Q. Du, "Discriminative reconstruction constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4666–4679, Jul. 2020.

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[58] B. Sun, Z. Zhao, D. Liu, X. Gao, and T. Yu, "Tensor decomposition-inspired convolutional autoencoders for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4990–5000, 2022.
[59] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1037–1050, Mar. 2022.
[60] K. Tan, Z. Hou, F. Wu, Q. Du, and Y. Chen, "Anomaly detection for hyperspectral imagery based on the regularized subspace method and collaborative representation," *Remote Sens.*, vol. 11, no. 11, p. 1318, Jun. 2019.
[61] C.-I. Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5131–5153, Jun. 2020.
[62] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[63] S. Chang and P. Ghamisi, "Nonnegative-constrained joint collaborative representation with union dictionary for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5534913.
[64] P. Xiang, S. Ali, S. K. Jung, and H. Zhou, "Hyperspectral anomaly detection with guided autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5538818.
[65] M. Wang, Q. Wang, D. Hong, S. K. Roy, and J. Chanussot, "Learning tensor low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 679–691, Jan. 2023.
[66] H. Qin, Q. Shen, H. Zeng, Y. Chen, and G. Lu, "Generalized nonconvex low-rank tensor representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5526612.
[67] T. Guo, L. He, F. Luo, X. Gong, Y. Li, and L. Zhang, "Anomaly detection of hyperspectral image with hierarchical antinoise mutual-incoherence-induced low-rank representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510213.
[68] N. Huyan, X. Zhang, D. Quan, J. Chanussot, and L. Jiao, "Cluster-memory augmented deep autoencoder via optimal transportation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531916.
[69] Z. Wu and B. Wang, "Transformer-based autoencoder framework for nonlinear hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508015.



Yu Huo received the B.E. degree in smart grid information engineering and the M.S. degree in control science and engineering from the Zhengzhou University of Light Industry, Zhengzhou, China, in 2019 and 2022, respectively. He is currently pursuing the Ph.D. degree in instrument science and technology with the School of Aerospace Science and Technology, Xidian University, Xi'an, China.

His research interests include hyperspectral anomaly detection, hyperspectral image classification, and object detection.



Xi Cheng (Student Member, IEEE) received the M.S. degree in control theory and control engineering from the Zhengzhou University of Light Industry, Zhengzhou, China, in 2021. He is currently pursuing the Ph.D. degree in instrument science and technology with the School of Aerospace Science and Technology, Xidian University, Xi'an, China.

His research interests include hyperspectral anomaly detection and saliency object detection.



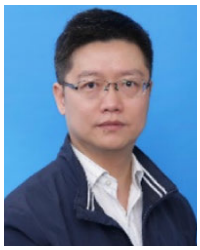
Sheng Lin received the B.E. degree in electrical engineering and automation and the M.S. degree in electrical engineering from the Zhengzhou University of Light Industry, Zhengzhou, China, in 2017 and 2020, respectively. He is currently pursuing the Ph.D. degree in instrument science and technology with the School of Aerospace Science and Technology, Xidian University, Xi'an, China.

His research interests include hyperspectral anomaly detection and object detection.



Min Zhang (Member, IEEE) received the B.E. and Ph.D. degrees in measurement and control from Xidian University, Xi'an, China, in 2010 and 2017, respectively.

She is currently an Associate Professor with the School of Aerospace Science and Technology, Xidian University. Her research interests include time and frequency measurements, digital signal processing, object detection, and hyperspectral image processing.



Hai Wang received the M.S. degree in communication and information systems and the Ph.D. degree in instrument science and technology from Xidian University, Xi'an, China, in 2003 and 2007, respectively.

He is currently a Professor with the School of Aerospace Science and Technology, Xidian University. His research interests include time and frequency measurements, frequency and phase measurements methods based on field-programmable gates array and system-on-chip, electronic instrument design, time–frequency signal processing, object detection, and hyperspectral image processing.

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